Air Quality, Knowledge Worker Performance, and Adaptation: Evidence from GitHub*

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Abstract

Highly skilled knowledge workers are important drivers of innovation and growth. We study how air pollution affects these workers, using data from GitHub, the world's largest coding platform, on 34,000 software developers across five continents. Pollution shocks reduce daily output quantity and quality by 4.3% and 2%, respectively, implying annual losses of $266 per developer. Developers adapt by working on easier tasks and by reallocating work to weekends following high-pollution days. We exploit location changes among developers to assess the effect of prolonged pollution exposure, which we estimate to be seven times as large as the effect of acute exposure.

JEL Codes: D24, J22, J24, L86, Q52, Q53
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1 Introduction

Environmental shocks generate major economic costs. Extensive research has shown that air pollution and extreme temperatures negatively affect human health, cognitive performance, and labor productivity (e.g. Deryugina et al., 2019; Ebenstein et al., 2016). However, while there is growing research on the effects of environmental conditions on workers and firms (e.g. Adhvaryu et al., 2022; Somanathan et al., 2021), little is known about impacts on high-skilled workers in the modern knowledge economy. Filling this gap is important, because knowledge work has grown rapidly, and is now a major driver of economic growth and competitiveness. Over the last decades, computerization has been causing an increase in the demand for workers performing non-routine, analytical, and interpersonal tasks (Autor and Price, 2013). Especially skills that complement digital technologies have been growing in importance. In the US, the share of jobs requiring intensive digital skills has more than quadrupled, increasing from 5% to 23% between 2002 and 2016 (Muro et al., 2017).[1]

In this paper, we study how air pollution impacts performance among highly skilled knowledge workers in a modern work environment and investigate adaptation responses. Air pollution is a ubiquitous environmental issue, with more than 90% of the global population being exposed to levels of fine particulate matter that exceed World Health Organization (WHO) guidelines. It has been shown to reduce performance across several settings, but existing research has considered routine jobs or inflexible work contexts (e.g. Chang et al., 2019; Archsmith et al., 2018). Knowledge workers not only employ different skills, but they also have more discretion in organizing their workday, with flexible schedules and task autonomy possibly enabling them to adapt to productivity shocks and to alleviate output effects. Moreover, these workers typically work in collaborative settings, where impacts of environmental shocks might get dampened, e.g., if co-workers can help each other, or get amplified due to complementarities if co-workers rely on each other's input. Since jobs characterized by these task profiles, skill requirements, and organizational features form the backbone of today's knowledge economy and are expected to become even more important as digitization and automation proceed, understanding how workers in these jobs respond to environmental shocks is essential to assess the economic cost of these shocks and to design optimal environmental policies.

To address the challenge that the output of knowledge workers is often difficult to observe, we use high-frequency data from *GitHub*, allowing us to measure output and work patterns of professional software developers. GitHub, the world's largest code hosting platform, is used for storing coding projects and puts great emphasis on facilitating collaboration between developers. As a STEM (science, technology, engineering, and math) occupation, software development requires analytical and advanced digital skills and generates high value for consumers, other

[1] The rapid adoption of generative artificial intelligence so far seems to further intensify this trend (OECD, 2026).

industries, governments, and the research community.[2] Therefore, adverse productivity effects in this occupation would have important implications for growth, innovation, and competitiveness. Moreover, software developers often work in flexible settings that offer discretion over working hours and task selection. These features make software development on GitHub a representative setting for modern knowledge work.[3]

We construct a user-by-day panel including measures of work quantity and quality, working hours, and task choice for 34,000 users across 59 countries covering the period between January 2014 and May 2019. We focus on users who work on projects owned by tech companies, indicating that they are professional software developers. Based on developers' locations, we match these outcomes to city-level air quality monitor data on particulate matter smaller than 2.5 µm ($PM_{2.5}$). To account for endogeneity in air quality, we follow previous research (e.g. Arceo-Gomez et al., 2016) by instrumenting $PM_{2.5}$ concentration with local temperature inversions. The instrumental variable (IV) strategy exploits the effect of plausibly exogenous changes in vertical temperature profiles in the atmosphere on local air pollution concentrations, controlling for a wide range of other weather characteristics. In addition to the analysis of acute pollution exposure, we investigate effects of *prolonged* exposure in a sample of 3,194 developers who change location during the observation period. If the effects of air pollution accumulate over time, prolonged exposure might have a larger effect than what would be implied by just adding up the estimated effect of repeated acute exposure. Using an inverse movers design similar to Lubczyk and Waldinger (2025), we compare post-move productivity between developers who moved to the same location at the same point in time, but from different origin cities in the same country, where they were exposed to different long-run average $PM_{2.5}$ concentrations.

We present three main findings. First, both the quantity and quality of software developers' output falls significantly if same-day air pollution increases. For a unit increase in $PM_{2.5}$ concentration, the number of daily actions observed on GitHub falls by 0.08%, and the acceptance rate of code changes a developer proposes drops by 0.06%, indicating a higher frequency of errors. When $PM_{2.5}$ reaches unusually high levels of at least one standard deviation above the city-by-month-by-day of week-specific average, the reductions amount to 4.3%, and 2%, respectively. Back-of-the-envelope calculations indicate that permanently improving air quality by 1 $\mu g/m^3$ would generate annual productivity benefits of $54 per developer, a relevant magnitude compared to reduced mortality and medical cost of $117 per elderly individual (Deryugina et al., 2019). Eliminating unusually high pollution days would generate benefits of

[2]Median annual pay of software developers in the US was $110,140 in 2020 (Bureau of Labor Statistics, 2021).

[3]Evidence in Appendix Figures B.1 and B.2 shows that software development resembles other high-skilled occupations in skill requirements like judgement and decision making (except for a greater need for digital skills), and features like work autonomy and frequent teamwork.

as much as $266 per developer and year. Extrapolating this estimate to similar STEM workers implies annual benefits of $1.24b per year in the US alone.[4]

Second, software developers adjust work patterns to adapt to the transitory pollution-induced productivity shock, exploiting their flexibility in task choice and schedules. On days with unusually high $PM_{2.5}$ levels, code that developers submit or review changes 3.4% fewer files and contains 4.2% fewer new lines compared to low pollution days, indicating that developers address less complex tasks. The least experienced developers show the strongest shift towards easier tasks in response to $PM_{2.5}$ exposure, attenuating effects on the number of actions they complete. Software developers further adapt by reallocating work from high-pollution, low-productivity weekdays to low-pollution, high-productivity weekends. On days with unusually high $PM_{2.5}$ concentration, they quit working 5.1 minutes earlier, on average. In compensation, they extend work time by roughly the same amount on the following weekends for every day with unusually high pollution levels during the workweek, if air pollution on the weekend is moderate. Results indicate that they use the extra work time on weekends to tackle more complex tasks, making up for the adjustment on high-pollution days. The ability of knowledge workers to adjust work patterns likely explains, why the effect of acute pollution exposure is moderate compared to other occupations studied in previous research, including both physically and cognitively demanding jobs. However, the monetary losses are economically relevant because of the high value created. Moreover, adjustments like shifting to weekend work imply an additional welfare cost due to forgone leisure time and potential adverse impacts on work-life balance.

Third, exploiting location changes among developers, we find that developers who have been exposed to higher $PM_{2.5}$ concentrations in their origin city have lower activity levels post move than developers who arrived from less polluted cities in the same country. The estimates indicate that the adverse effect of long-term exposure to one additional unit of $PM_{2.5}$ on productivity is seven times as large as the effect of short-term exposure, consistent with accumulation of the effect of repeated exposure, and less scope for adaptation to high air pollution over prolonged periods. The negative effect of long-term exposure persists for three quarters after the move. This finding implies that just adding up the estimated effect of repeated acute exposure likely underestimates the potential economic benefit of policies that improve air quality permanently.

In additional analyses, we show that the adverse effects of pollution on output quantity arise at concentrations below current regulatory standards in the European Union and the US. Moreover, we find that the effects are more pronounced in locations with an older building stock, and thus likely higher indoor penetration rates, and in countries with low air pollution awareness. This points to a physiological mechanism and suggests that, apart from policies aimed at reducing air pollution, improving building infrastructure and air quality awareness

[4]Similar STEM workers refers to 4,654,750 workers in the occupation group 'Computer and Mathematical Workers' in the US in 2021 (Bureau of Labor Statistics, 2022).

may yield productivity gains and make the local economy more competitive. By contrast, we find no heterogeneity in effects by local income levels. We also analyse effect heterogeneity by type of work, showing that the overall drop in output on high-pollution days is mainly driven by a decline in *individual* coding activity and work on new tasks, whereas *collaborative and responsive* work (e.g., commenting on issues) is less affected.

Our paper makes several contributions to the literature. First, our work demonstrates new ways to use publicly available data on GitHub activity. While we are not the first to use this data in economics (for example, McDermott and Hansen (2021) analyze the impacts of the COVID-19 pandemic on GitHub work patterns), we propose strategies to construct a sample of highly active users who are likely professional software developers, to study task difficulty, and to estimate the monetary value of the output observed on GitHub.

Second, we link to the literature on air pollution and worker productivity. Several studies document a negative impact of pollution on productivity in manual and routine jobs, such as textile workers or call center agents (Adhvaryu et al., 2022; Chang et al., 2016, 2019; Graff Zivin and Neidell, 2012; He et al., 2019; D'Adda et al., 2025). A few papers also analyze effects of poor air quality on performance in more cognitively demanding work settings, including error detection by baseball umpires (Archsmith et al., 2018), quality of speeches held by politicians (Heyes et al., 2019), and case handling time by judges (Kahn and Li, 2020; Sarmiento, 2022). Related work investigates performance in cognitively demanding tasks outside of work settings, e.g., among chess or brain game players (Huang et al., 2020; Krebs and Luechinger, 2024; Künn et al., 2023; La Nauze and Severnini, 2025). While these settings provide precise performance measures in a specific domain, they do not capture the typical features of work organization in modern high-skill jobs. We expand the analysis to a STEM profession that is representative for a large group of knowledge workers in modern, flexible, and collaborative work environments. Thereby, our study adds novel insights into the labor market cost of air pollution in the context of ongoing digitalization.[5] Furthermore, our study, unlike others that often rely on data from a single country or location, uses a large, international sample of developers. This allows us to draw more general conclusions about the pollution-productivity relationship and to explore effect heterogeneity with respect to local income levels, building stock quality or pollution awareness, with relevant implications for policy.

Third, the broad international coverage and the fact that the we can follow software developers as they change locations also allow us to estimate effects of prolonged exposure to air pollution, contributing to a much scarcer literature strand, especially for labor productivity.[6] While Lubczyk and Waldinger (2025) investigate effects of sustained exposure to high pollution

[5] Borgschulte et al. (2024) and Fu et al. (2021) analyze pollution impacts on US labor earnings and Chinese manufacturing productivity, without focusing on specific professions.

[6] Anderson (2020) studies effects of long-run exposure to air pollution on health, exploiting pre-dominant wind patterns near highways in Los Angeles. Chen et al. (2013) and Ebenstein et al. (2017) exploit the discontinuity in air quality at the Huai river, and find a substantive loss in life expectancy due to higher long-run exposure to PM10.

levels in Socialist East Germany on labor market outcomes decades later, we complement this evidence by focusing on more subtle effects at lower baseline pollution levels that might not materialize in wages, but still imply a relevant economic cost.

Finally, we contribute to the literature on worker adaptation to environmental shocks and connect it to research on the effect of flexible work arrangements on productivity. Several papers study how workers adjust working hours in response to extreme temperatures (Graff Zivin and Neidell, 2014; Neidell et al., 2021; LoPalo, 2023). Concerning air pollution shocks, Adhvaryu et al. (2022) and Bassi et al. (2021) demonstrate how managers mitigate productivity losses, for example by reallocating workers to different tasks. In related work, Hoffmann and Rud (2024) find that informal workers in Mexico City reallocate labor supply across days to avoid extreme pollution exposure. While these studies focus on low-skilled workers in high pollution contexts, our work identifies new margins of adjustment in a flexible high-skilled setting, namely task choice and temporal reallocation of work activity towards the weekend, and covers a wide range of air quality concentrations. By showing that workers exploit their flexibility to adapt to a transitory productivity shock, and thereby alleviate its adverse impact, our work links to research on the causal effects of flexible work arrangements and worker autonomy on productivity (e.g. Beckmann et al., 2017; Angelici and Profeta, 2023). Our results suggest that the ability to adapt to idiosyncratic productivity shocks might contribute to the positive relationship between flexible work arrangements and performance.

2 Background

Particulate Matter. We focus on $PM_{2.5}$, particulate matter with a diameter of less than 2.5 µm, which encompasses all solid and liquid particles suspended in the air. In urban areas, most $PM_{2.5}$ originates from anthropogenic sources, including traffic, industrial production, and biomass burning (Karagulian et al., 2015). The main reasons to focus on $PM_{2.5}$ are (i) its ability to penetrate indoors, making it particularly relevant for indoor office workers (Deng et al., 2017), and (ii) the fact that a large body of research documents that fine particulate matter plays a key role for the adverse health effects of air pollution. The small particles can penetrate deep into the lungs, causing damage to the respiratory system, as well as cardiovascular health effects like high blood pressure and heart diseases (Lederer et al., 2021). While severe health damages are concentrated among vulnerable groups, even healthy adults can experience mild symptoms, including irritation in the nose and throat or coughing (Pope, 2000). Recent clinical and autopsy studies suggest that exposure to fine particles can even affect the central nervous system as the small particles can reach the brain, causing neuro-inflammation (Babadjouni et al., 2017). In line with this, higher $PM_{2.5}$ concentrations have been found to lower scores in online brain games (La Nauze and Severnini, 2025) and high-stakes exams (Ebenstein et al., 2016). In response to the evidence on adverse health effects, several countries introduced standards on annual ambient $PM_{2.5}$ concentrations and often tightened them over time. Currently, standards

are in place for example in the US (9 $\mu g/m^3$) and the European Union (25 $\mu g/m^3$). The WHO recommends a level of no more than 5 $\mu g/m^3$.

Open-Source Software Development. As our data only contain information about work done in public GitHub projects, our analysis is limited to open-source software (OSS) projects. OSS projects operate under licenses that allow anyone to access, modify, and distribute the source code freely. This may raise the concern to what extent OSS development is representative of other work done by programmers and how relevant it is for firms' economic success and the broader economy.

Prominent examples of OSS projects include Linux, Apache, Visual Studio Code, Python, or TensorFlow, which are all widely used across consumer and business applications. For example, as of March 2024, Apache servers powered approximately 20% of active websites, and Linux served as the operating system for 90% of cloud computing services and 99% of supercomputers (Netcraft, 2024; Corbet and Kroah-Hartman, 2017). Estimates by Hoffmann et al. (2024) suggest that the labor costs required to replicate widely-used OSS projects amount to $4.15 billion. Considering the actual usages by many economic agents implies demand-side values of OSS as large as $8.8 trillion.

Corporations, such as Microsoft and Apple, do not only contribute to many OSS projects, but have made part of their work open source by releasing technologies like .NET, ChakraCore, and Swift under open licenses (Landwerth, 2014; Apple Inc., 2024). In general, professional developers are key in mantaining many OSS projects. According to Riehle et al. (2014), around half of all open source development has been paid for by firms. At Alphabet for example, 10% of employees work regularly on OSS projects (Vargas, 2021). Andersen-Gott et al. (2012) grouped the motivation of firms to pay workers for OSS development into three categories: selling complimentary closed software products, building greater innovative capability, and cost reduction through open-sourcing to an external community. Other economic benefits from providing OSS are quickly receiving feedback from customers (Landwerth, 2014) and hedging against losing a software's original developer team (Raymond, 2001).

Furthermore, open source programming practices are increasingly adopted by commercial projects. Research by Kalliamvakou et al. (2015) suggests a convergence between open and closed-source projects, with commercial projects increasingly imitating the decentralized and autonomous workflows found in OSS projects.

3 Setting and Data

This section starts with a description of GitHub, followed by an overview of the GitHub data and how we use it to measure developers' productivity. After checking the validity of these outcome measures, we describe the environmental data.

3.1 Git and GitHub

GitHub is built on *Git*, an open-source version control system that tracks changes made to (code) files, recording who made the change and when. *GitHub* is a web platform for hosting Git repositories, the directories where all files belonging to specific projects are stored. On top of the version control functionality, GitHub also provides collaboration tools. For each repository (or repo for short), the owner can choose whether to make it public (visible to everyone) or private (visible only to repository members). In 2019, GitHub had more than 30 million registered accounts owning more than 120 million public repositories, making it the world's largest host of source code.

The core action in Git is a *commit*, which refers to saving the current state of the repository after making changes to one or more files. As such, a commit represents that coding work was conducted by the commit author. Only repository owners and invited team members can modify files through commits.

The primary collaboration features offered on GitHub are *pull requests* and *issues*. A pull request (PR) is used to propose code changes to a repo. To create a PR, a user creates a copy of the repository, implements changes in their copy via commits, and then submits these to the original repository. Repo members then review the proposed changes and decide whether to accept (i.e., merge) or reject them. Each PR includes a discussion forum where users can comment directly on the proposed changes to provide feedback. Hence, PRs facilitate collaborative coding and are used not only for external contributions but also within project teams.

Issues are text messages typically used to organize tasks within a repo. Like PRs, issues contain a discussion forum where users can comment on the matter at hand. Repository members can assign labels to issues to highlight their category (e.g., bug, feature request), priority, or difficulty. GitHub provides nine default labels, and teams can create additional labels specific to their repo. Once an issue is resolved, it can be marked as closed.

Additionally, GitHub has social network functions, e.g., options to follow other users and subscribe to specific repos and issues to receive notifications about new activities.

Since GitHub actions related to commits, PRs, and issues reflect work aimed at building or improving software products, we can use data on these activities to measure output of highly skilled developers.

3.2 GitHub Data on Productivity and Work Patterns

The *GHTorrent* project provides a database on GitHub users and all their actions in *public* repositories. We use the version of the database containing data up to June 1st, 2019. The *user* table comprises a unique identifier, login name and registration date for all users registered by that date. In addition, location and company information as stated on the user profile on this date is reported. The *projects* table provides identifiers and names of all public repositories as

well as a reference to the user owning the repo. Activity data is available by type (e.g., commits, opening issues, PR comments), including exact timestamps and the identifiers of the acting user and the repository where the event occurred. For certain actions, further information is reported, such as labels attached to issues. We complement this with data from *GHArchive*, which also records actions in public repositories and contains additional information on some events, e.g., the number of lines of code added and deleted, and the number of files changed within a PR. GHArchive and GHTorrent data can be linked via users' login names.

These data have several favorable features for our analysis. First, the precise records of GitHub activities allow us to quantify the daily output of software developers, addressing the long-standing challenge of measuring the work of highly skilled workers during a given period. Second, the data cover all GitHub users, providing a broader geographic coverage and thus an advantage in terms of external validity compared to previous studies. Moreover, the rich information included allows us to measure not only output quantity but also quality and work patterns, which are crucial in knowledge work.

The data also have limitations. To assign local air quality to users, we rely on self-reported locations, which may be wrong or outdated, giving rise to measurement error. This, under the assumption of classical measurement error, leads to attenuation bias such that any adverse effect found represents a lower bound. Additionally, we lack information on work conducted in private repositories or outside GitHub. Many users engage in no or minimal work in public repositories, making it impossible to measure their productivity based on their activity data. Thus, when constructing our analysis sample, we aim to capture users who are professional software developers and do a substantial part of their formal work in public GitHub repos.

Sample Construction. We focus on users who report a location at the city level so that we can assign local air quality. From this group, we only keep users who have ever committed in a repository owned by a company, i.e., users with the authority to change the source code of a company-owned project. This step is intended to focus on professionals who are affiliated with the companies. To identify these users, we compile a list of company-operated repositories[7]. To exclude bots—computer programs often used to automate routine tasks—we discard a small number of users with bot-like behavior.[8]

To focus on cases where we observe a substantial part of an individual's total work, we admit users to the sample once they made at least 20 commits in public repos in a given month. They enter the sample in the month after meeting this threshold for the first time. Users remain in the sample until the end of the observation period unless they conduct fewer than three *unproductive* actions in a given month. In such cases, we drop users from the sample for that month, assuming they moved to a different platform or private repos. Unproductive actions are

[7]This list is based on https://github.com/d2s/companies/blob/master/src/index.md and lists of open-source projects operated by Google, Microsoft and Facebook mentioned on their websites.

[8]We exclude users in the top 0.1 activity percentile, with over 20% of commits occurring on exact full hours, or with bot-indicative login names.

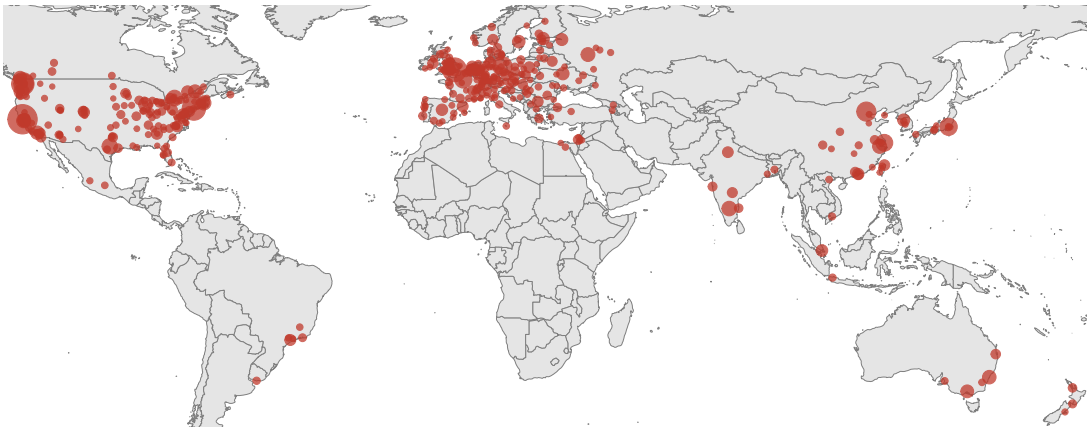


Figure 1: Sample Cities

Notes: Points represent sample cities. Circle size is based on the number of users observed in the city.

not used as outcomes and are based primarily on the social network functions GitHub offers, including following other users, watching repositories, (un)subscribing to issues, labeling issues, and (un)assigning issues to users. Lastly, we restrict the sample to users living in cities with at least 5 relevant users and covered by our air pollution data. This yields 34,135 users across 489 cities in 59 countries (Figure 1).

Outcome Measures. We compile an unbalanced user-by-day panel for the period between January 2014 and May 2019, including measures of output quantity, quality, and work patterns.[9] To measure daily work *quantity*, we count the total number of productive actions conducted, given as the sum of commits, PR and issue creations, PR and issue closings, issue reopenings, and comments on PRs, issues or commits. Furthermore, we separately count commits, as a measure of individual coding work, and comments, as a measure of interactive activity in response to work conducted by others, to account for potential effect heterogeneity between activity categories. To assess output *quality*, we measure the percentage of PRs opened on a given day that are merged, indicating acceptance, and thus satisfactory code quality.

To assess worker *adaptation*, we build measures of task choice and working hours. Firstly, to explore whether users switch to easier tasks on high-pollution days, we consider the complexity of issues and PRs. For PRs, we count the number of new lines of code added, lines of code deleted, and code files changed. We take the average value of these variables across all PRs a user worked on a given day, either by creating or reviewing the respective PR. For issue difficulty, we rely on the user-assigned issue labels which indicate that a given issue is relatively easy, e.g., the default labels *good first issue* and *documentation* (Tan et al., 2020), or individual labels such as *low-hanging fruit*. The complete list of labels we use to identify easy tasks is depicted in Appendix Table A.1. We construct the share of all issue events conducted by

[9]During our sample period some users changed their location. Since the GHTorrent data on users is a snapshot taken on June 1st, 2019, we use earlier versions of the database (one snapshot in each year between 2015 and 2018) to check for movements. 6.3% of users reported more than one distinct location. We identify the city where they spent the biggest part of the sample period, and keep them in the sample only while residing in this city.

Table 1: Summary Statistics for the Analysis Sample of GitHub Users

	Mean	SD	SD (within)	Min	Max	Observations
Output Quantity						
Actions	2.521	6.857	6.117	0	291	19,576,311
of which Commits	1.18	3.591	3.342	0	194	19,576,311
Comments	0.848	3.224	2.831	0	280	19,576,311
PRs opened	0.135	0.668	0.626	0	190	19,576,311
Issues opened	0.099	0.764	0.715	0	206	19,576,311
PRs closed	0.147	0.803	0.755	0	142	19,576,311
Issues closed	0.107	0.829	0.805	0	263	19,576,311
Any action	0.345	0.475	0.431	0	1	19,576,311
Actions | Actions > 0	7.302	10.064	8.961	1	291	6,758,242
Output Quality						
PR acceptance rate	0.664	0.455	0.402	0	1	1,382,964
Task Complexity						
Share easy issue events	0.029	0.139	0.134	0	1	4,015,135
Files changed per PR	8.351	52.501	51.329	0	7951	1,989,189
Lines added per PR	269.296	1264.51	1229.952	0	36907	1,989,189
Lines deleted per PR	106.695	573.787	561.028	0	17277	1,989,189
Working Hours						
Time first action	13:06	4.96 hours	4.56 hours	3:00	3:00	6,732,524
Time last action	17:29	5.05 hours	4.67 hours	3:00	3:00	6,732,524

Notes: Table shows key measures of output quantity, quality, complexity, and working hours at the developer×date level.

the user (commenting, opening, closing, or reopening of issues) which refer to an *easy* issue. Secondly, we investigate whether users adjust work time in response to pollution. We measure the time of the first and last action of the day (in minutes since midnight). At the user-by-week level, we further measure the extent of weekend work by the average time of the first and last action, respectively, on the weekend.

All outcomes, with details on their construction, are listed in Appendix Table A.2.

Descriptives. While our sample comprises just 0.15% of all GitHub users, they conduct a disproportionally large share of activities in public repositories, namely 5.8% of issue creations, 9% of issue closings, 12% of comments, and 8% of PR openings and closings. Hence, the selected users are a highly active sample. This is confirmed by the summary statistics on the outcome variables (Table 1). The average number of daily actions is 2.52, of which 1.18 are commits and 0.85 are comments. The remaining productive GitHub actions related to issues and PRs occur less frequently. On average, users are active on 35% of all days, and conditional on being active, the mean number of actions is 7.3. The share of PRs getting accepted is 66%. In the average PR, more than 8 distinct files are changed and 269 new lines of code are added. On average, 2.9% of all issue events (opening, closing, reopening, or commenting on an issue) refer to an easy issue. The mean time of the first and final action of the day are 1:06 pm and 5:29

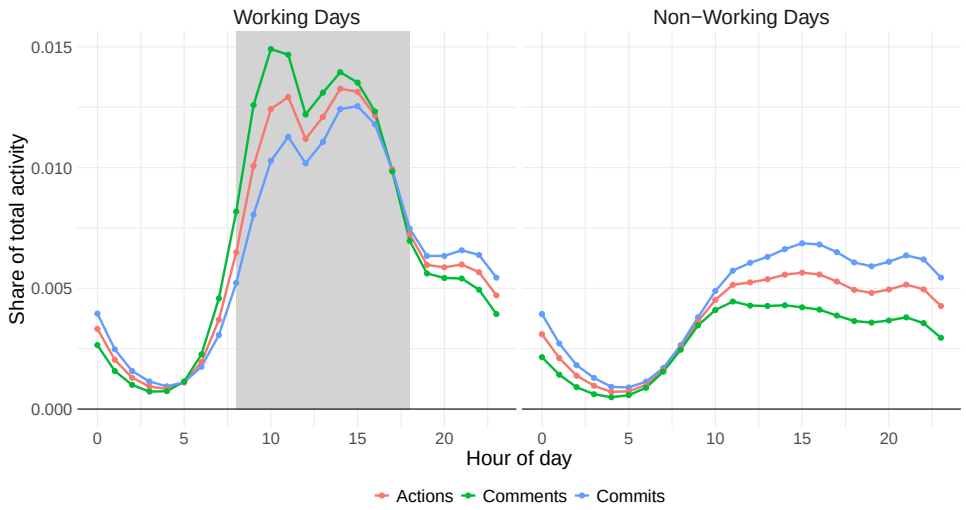


Figure 2: Distribution of Activity across Hours of the Day and Days of the Week

Notes: Hourly share of activities on working days (left) or non-working days (right, weekends and public holidays), for total actions, comments, and commits. Grey area: core working hours, 8 am to 6 pm on weekdays.

pm, respectively.[10] Appendix Table A.3 depicts summary statistics by type of day, showing that the average number of total actions is 3.0 on working days, more than twice as high as on weekends and holidays (1.4).

Figure 2 depicts the distribution of activity across days of the week and hours of the day. The lines represent the share of all activity conducted during each hour of the day on weekdays (left) or weekends and public holidays (right) for commits, comments, and total actions. Activity levels peak during core working hours and decline in the evening and night hours and on non-working days. Notable activity during evening hours and on weekends is not uncommon among highly educated workers (Mas and Pallais, 2020). Comments, i.e., more interactive activities, are even more concentrated during standard working hours compared to commits, i.e., individual coding activities. This is plausible as more collaborative tasks are more productive at times when other users are working as well.

Finally, Figure 3 presents information about the work status of users in our sample. The left plot depicts the most frequent terms used in the biographies on their GitHub profiles, after stemming and removing stop words. 30% of the users provide such a self-description. Three terms stand out: *engineer/ing*, *software*, and *developer/development*. The right plot complements this with information on employers. 64% of the users provide employer information on their profiles. The most frequently reported employers are major tech companies strongly engaged in open-source. The high prevalence of work-related terms and notable tech firms, along with the peak in activity during core working hours, strongly suggests that we are capturing

[10]To account for the tendency of high-skill workers to work long hours in the evening, we define a workday from 3 am on the calendar date to 3 am on the following day. Note that the time of the first recorded action reflects the point in time when the first action is completed, e.g., the first code chunk is written and committed.

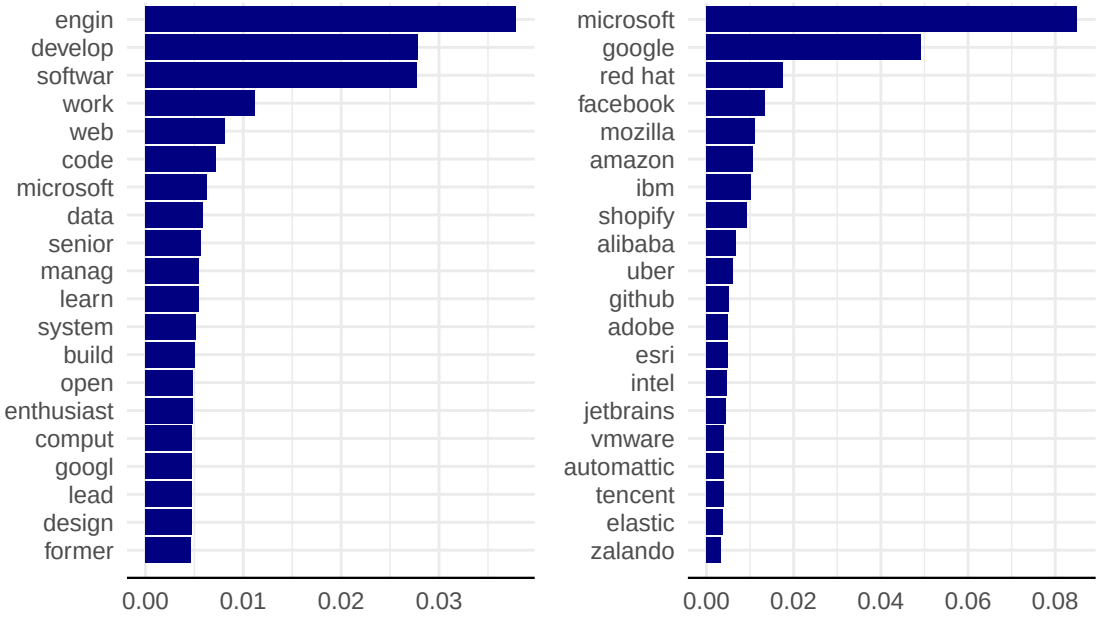


Figure 3: Most Frequent Terms from User Self-Descriptions and Company Fields

Notes: Left: Most frequent terms from user self-descriptions, after stemming and removing stopwords. Data from 10,340 user bios, accessed in 2021 via the GitHub API. Right: Most frequent companies. Data from June 2019 on 21,940 users. Horizontal axes measure share of users reporting the respective term or company.

professional developers using GitHub for formal work. Thus, in the remainder of the paper, we refer to the sample users as 'developers'.

3.3 Gitcoin: Monetary Value of GitHub Activity

To validate our performance metrics and to translate the effects of air pollution into monetary damages, we draw on data from the web platform *Gitcoin*. On Gitcoin, GitHub project teams aiming to incentivize external contributions to their projects post issues from their public repos along with a payment they offer for a solution. Freelance developers can apply to solve these issues. Work on the issues is submitted in the form of a PR in the respective GitHub repo. If the PR is accepted by the issue funder, the PR author receives the payment, typically in cryptocurrencies.

We collected data on 292 issues for which PRs were submitted and payments made by March 2022 via the Gitcoin API, including the value of the payment in USD and the hours worked on the PR as reported by the submitting user. We merge this with information on the size of the respective pull request, obtained via the GitHub API (number of commits, number of lines of code added and deleted, and number of files changed). A detailed description of the data is provided in Appendix C.

We find mean payments of $354 per pull request and $112 per commit. Developers spend on average 1.8 hours per commit. This implies an hourly wage of $62, very close to the mean

wage of $58 among software developers in the US in 2021 (Bureau of Labor Statistics, 2021). We will use the monetary values of commits and PRs to translate the effects of air pollution into monetary damages.

Are the outcomes we consider valid measures of productivity and task complexity? In Appendix Tables C.1 to C.3 we use the Gitcoin data to test this. Both, payments awarded for PRs and time spent on PRs, are positively correlated with the number of commits that PRs comprise. This confirms that changes in the number of commits reflect fluctuations in developer productivity. Additionally, holding the number of commits constant, adding more lines of code and changing more files in a PR is associated with a higher payment, suggesting that these variables indeed reflect task complexity. 12% of the Gitcoin issues are labeled as *easy* according to our definition. Even when controlling for all previously mentioned PR characteristics, these issue labels are associated with a 30% lower payment for a PR, supporting their validity as indicators for easy tasks.

3.4 Environmental Data

Air Quality. We collect $PM_{2.5}$ concentration data from outdoor monitors from several environmental agencies. For cities without publicly available monitor data, we use high-resolution reanalysis data from the Copernicus Atmosphere Monitoring Service (CAMS). Reanalysis data are constructed by combining ground-level measurements, satellite images, and atmospheric transport models. Appendix Table A.4 provides a list of the data sources. The data are provided at daily or hourly intervals. We aggregate hourly data to the daily level and assign cities the simple average of all monitor readings within a 40 km radius around the city centroid. For cities covered by the CAMS data, we only use points within 25 km due to the high resolution. Our data on $PM_{2.5}$ covers 98% of observations in the GitHub panel.

We winsorize $PM_{2.5}$ at the continent-specific 0.1th and 99.9th percentile to mitigate the influence of extreme outliers, such as heavy wildfire smoke or industrial accidents. The population-weighted average $PM_{2.5}$ concentration in the sample is 15.05 $\mu g/m^3$ (standard deviation: 22.0 $\mu g/m^3$, within-city: 16.4 $\mu g/m^3$), which is between the regulatory standards on annual $PM_{2.5}$ concentration in the US (9 $\mu g/m^3$) and the EU (25 $\mu g/m^3$). Figure 4 displays the distribution of daily $PM_{2.5}$ for eleven large geographic regions, $R \in$ {Northern Europe, Southern Europe, Western Europe, Eastern Europe, North America, South America, Oceania, Eastern Asia, Western Asia, Southern Asia, South-Eastern Asia} and demonstrates substantial regional heterogeneity. While concentrations above 20 $\mu g/m^3$ are infrequent in North America, Oceania, and Northern Europe, they are the norm in Southern and Eastern Asia, and occur on close to 20% of all days in cities in South America and Southern Europe.

Thermal Inversions. The instrumental variable approach is based on thermal inversions, which are defined as atmospheric conditions where the temperature of upper air layers is *higher* than the temperature at the earth's surface. Under normal conditions, air temperature

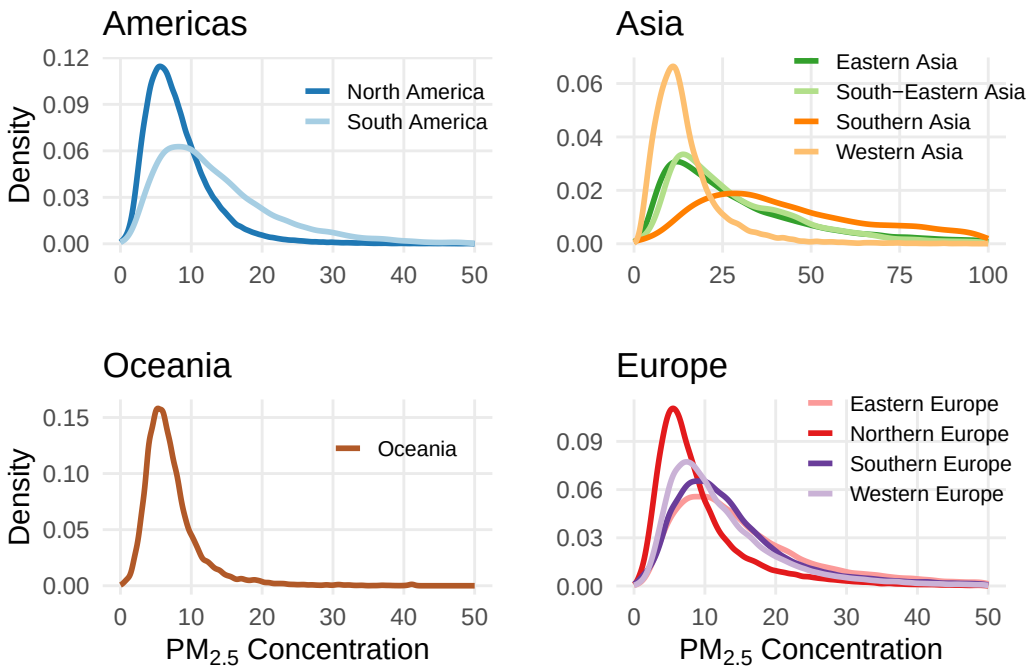


Figure 4: Regional distribution of daily $PM_{2.5}$

Notes: North America: United States, Canada, Mexico. South America: Brazil, Uruguay. Oceania: Australia, New Zealand. Northern Europe: Denmark, Estonia, Finland, Ireland, Latvia, Lithuania, Norway, Sweden, United Kingdom. Southern Europe: Bosnia and Herzegovina, Croatia, Greece, Italy, Malta, North Macedonia, Portugal, Serbia, Slovenia, Spain. Western Europe: Austria, Belgium, France, Germany, Luxembourg, Netherlands, Switzerland. Eastern Europe: Belarus, Bulgaria, Czechia, Hungary, Moldova, Poland, Romania, Russia, Slovakia, Ukraine. Eastern Asia: China, Hong Kong, Japan, South Korea, Taiwan. Western Asia: Armenia, Egypt, Georgia, Israel, Jordan, Türkiye. South-Eastern Asia: Indonesia, Singapore, Vietnam. Southern Asia: Bangladesh, India.

decreases with altitude. To construct inversion measures, we collect reanalysis data on hourly temperature at the surface level and at several pressure levels from the European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 products. The data is reported on a 0.25-degree grid, corresponding to roughly 28 km × 28 km at the equator. For each grid point and hour, we compute the difference between upper air temperature and surface air temperature in degrees Celsius. Upper air temperature is measured at the pressure level 25 hPa above the surface pressure level, which corresponds to an altitude difference of approximately 210 meters.

We then calculate the average temperature difference during nighttime hours (midnight to 6 am), after translating timestamps into local time. To obtain city × day level variables, each city is assigned the inverse distance weighted average nighttime temperature difference from the four grid points closest to its centroid. This continuous measure of inversion strength will be our main instrument for local pollution. In robustness checks, we use an inversion indicator instead, which takes value one if the difference between upper air and surface temperature is positive and zero otherwise. Inversions occur on 45% of all days during the sample period, and the median temperature difference is -0.4°C, (standard deviation: 2.83, within-city: 2.66). Appendix Table A.5 shows the frequency of inversions by geographic region.

Further Meteorological Conditions. To construct controls for daily weather conditions we use the ERA5-land product from the ECMWF. It provides hourly data on air temperature, precipitation, and dewpoint temperature on a 0.1-degree grid. We compute daily mean temperature, precipitation, and relative humidity at each grid point and then assign each city the inverse distance weighted average weather conditions from the eight closest grid points.

We collect reanalysis data on wind conditions from the Japan Meteorological Agency's JRA-55 product. The u- and v-component of wind, i.e., the wind vectors, are reported every six hours on a 1.25-degree grid. We aggregate to the daily level and compute the inverse distance weighted average of u- and v-vectors at the four grid points located closest to each city's centroid. Finally, daily average wind speed is computed from the city-level u- and v-vectors.

Some of the largest cities in our sample are located along the North American west coast, frequently experiencing severe wildfires, which can lead to massive air pollution due to smoke. Since wildfire smoke can trigger avoidance behavior (Burke et al., 2022), we control for heavy smoke events to ensure that our estimates capture physiological impacts of $PM_{2.5}$ exposure rather than behavioral responses to wildfires. The National Oceanographic and Atmospheric Administration's Office of Satellite and Product Operations provides data on the location and intensity of smoke plumes across North America. We define a city as being affected by wildfire smoke if a plume overlaps with a 15 km radius around its centroid, and aggregate the data to the daily level by summing over the intensities of all plumes covering a city. We define a heavy smoke indicator taking value one if the city was covered by a plume of the highest intensity or if the sum exceeds twice the maximum intensity. 0.3% of all city-day observations and 9% of all observations with any smoke exposure are classified as heavy smoke days.

4 Research Design

Baseline Regression Model. To analyze how short-run variation in local particulate matter concentration affects professional software developers, we specify a model for developers living in city c on day d:

$$y_{c,d} = \beta PM_{c,d} + \boldsymbol{w}'_{c,d}\boldsymbol{\gamma}_{R(c)} + \delta_{R(c)}h_{c,d} + \mu_c + \mu_{R(c),yr(d),m(d)} + \mu_{R(c),dow(d)} + \mu_{cn(c),m(d)} + \varepsilon_{c,d}. \quad (1)$$

Here, $y_{c,d}$ denotes one of the measures of output quantity, quality, or work patterns described in the previous section. We obtain this variable through an auxiliary regression that controls for developer experience in using GitHub and a developer fixed effect. By doing so, we can reduce the computational burden without losing variation in the regressor of interest – particulate pollution $PM_{c,d}$ – which is observed at the city-day level. This procedure is commonly employed (e.g. Currie et al., 2015) and asymptotically equivalent to estimating the underlying individual-level regressions (Donald and Lang, 2007). Appendix D provides a more detailed description.

The fixed effect μ_c controls for time-invariant unobserved factors at the city level. Region-specific year × month fixed effects $\mu_{R(c),yr(d),m(d)}$ capture time-varying productivity shocks common to all developers in a geographic region R. Region-specific day-of-week fixed effects $\mu_{R(c),dow(d)}$ and an indicator for holidays, $h_{c,d}$, control for fluctuations in work patterns and productivity across days of the week and public holidays. These fluctuations are allowed to vary in intensity across different world regions. To control for location-specific seasonality in pollution and work patterns, we add country × calendar month fixed effects $\mu_{cn(c),m(d)}$. The vector $\mathbf{w}_{c,d}$ contains weather variables that can be correlated with air quality, inversions, and work patterns. It includes a series of indicator variables for daily mean temperature falling into bins defined based on the 5th, 10th, 20th, 35th, 65th, 80th, 90th, and 95th percentiles of the city-specific temperature distributions, cubic polynomials of precipitation, relative humidity, and wind speed, as well as a dummy indicating heavy wildfire smoke. The effects of all weather variables are allowed to differ across regions R, taking into account that for instance individuals in warmer regions might respond differently to unusually warm temperatures than individuals in colder regions. We weight all regressions by the number of developer observations in the city–day cell and cluster standard errors at the city level.

While we include a wide range of controls to account for sorting into different cities or fluctuations in economic conditions, $PM_{c,d}$ may still be endogenous in model (1) due to unobservable factors that co-vary with particulate matter and productivity. Variations in local economic conditions can for instance affect air pollution and developers' output at the same time. Similarly, local events like a football match or the closing of a bridge may impact both traffic and work patterns. Besides omitted variable bias, a second issue is measurement error in individual pollution exposure, which we have to proxy for by city-level averages, generating attenuation bias in the OLS estimator.

IV Estimation. To address endogeneity, we follow several recent studies (e.g. Arceo-Gomez et al., 2016) and instrument local pollution levels with temperature inversions. During an inversion period, air temperature increases with altitude, such that a warm upper air layer acts as a ceiling, preventing pollution emitted at the surface level from dispersing and trapping it near the ground. Consequently, the same amount of surface-level emissions results in higher pollution concentrations compared to normal periods. This effect on air quality is stronger, the stronger the inversion, i.e., the larger the difference between the upper and surface air temperature. Our instrument for pollution will therefore be the temperature difference between upper and lower layers, $Inv\ Strength_{c,d} = \Delta T_{c,d}$.

Following Jans et al. (2018) and Sager (2019), we focus on nighttime inversion strength, as daytime inversions may be visually noticeable in some regions which could lead to behavioral adjustments. Nighttime inversions are less likely to trigger such responses, strengthening the exogeneity of the instrument.

Inversions can be correlated with weather conditions. For instance, calm winds tend to reduce the mixing of cold and warm air, creating favorable conditions for the formation of inversions. Weather conditions could also affect labor-leisure trade-offs (Graff Zivin and Neidell, 2014) and thereby the output of developers via channels other than air quality. Moreover, in many places, inversions exhibit seasonality, with higher frequency during winter. Therefore, it is important to control for the wide range of weather conditions in $\mathbf{w}_{c,d}$ and to include fixed effects capturing seasonality effects.

In our global sample, the impact of a temperature inversion on air quality is not uniform across all cities. When a warm upper air layer hinders pollution from dispersing, this causes a strong increase in pollution concentration in locations with high emissions, whereas areas with low levels of ground-level emissions are much less affected (Krebs and Luechinger, 2024). Additionally, the effects of inversions can differ depending on geographic and topographic factors, such as when mountains further impede air circulation. To account for this, we allow the impact of inversion strength on $PM_{2.5}$ to vary between groups of cities that are geographically close to each other and exhibit similar pollution levels. We use a k-means clustering algorithm to assign each city into a group g based on longitude, latitude, and average $PM_{2.5}$ concentration during the sample period. In our baseline specification, we form 25 groups, which are depicted in Appendix Figure B.3.

The first stage of the IV estimation is as follows, where the coefficient π^g is allowed to vary across city-groups $g \in \{1, 2, \ldots, 25\}$:

$$PM_{c,d} = \pi^g \Delta T_{c,d} + \mathbf{w}'_{c,d}\boldsymbol{\gamma}_{R(c)} + \delta_{R(c)} h_{c,d} + \mu_c + \mu_{R(c),dow(d)} + \mu_{R(c),yr(d),m(d)} + \mu_{cn(c),m(d)} + \varepsilon_{c,d}. \quad (2)$$

Figure 5 illustrates the first stage relationship for five city groups. It shows binned scatter plots of residualized $PM_{2.5}$ concentration, after taking out the weather controls and fixed effects mentioned above, against inversion strength. A linear function fits the relationship between inversion strength and air pollution reasonably well. As anticipated, an increase in the temperature difference between upper and surface level air has a more pronounced effect on $PM_{2.5}$ concentration in cities with higher baseline pollution levels (e.g., Beijing vs. New York). In Appendix Figure B.4 we present these plots for all 25 city groups.

Pollution Measures. Our primary measure of air pollution is the daily average $PM_{2.5}$ concentration in $\mu g/m^3$. As an alternative measure, we define a binary variable that we will refer to as a *pollution shock*,

$$PM_{2.5}\ shock_{cd} \equiv \mathbf{1}\left\{PM_{c,d} > \sqrt{\widehat{\mathrm{Var}}\ [PM \mid c]} + \widehat{\mathrm{E}}\ [PM \mid c, m(d), dow(d)]\right\}. \quad (3)$$

The indicator takes value one if the city-day concentration of $PM_{2.5}$ is more than one city-specific standard deviation above the level expected for the given city, calendar month, and day of week. In essence, it reflects unusually high pollution levels in a location × season × day-

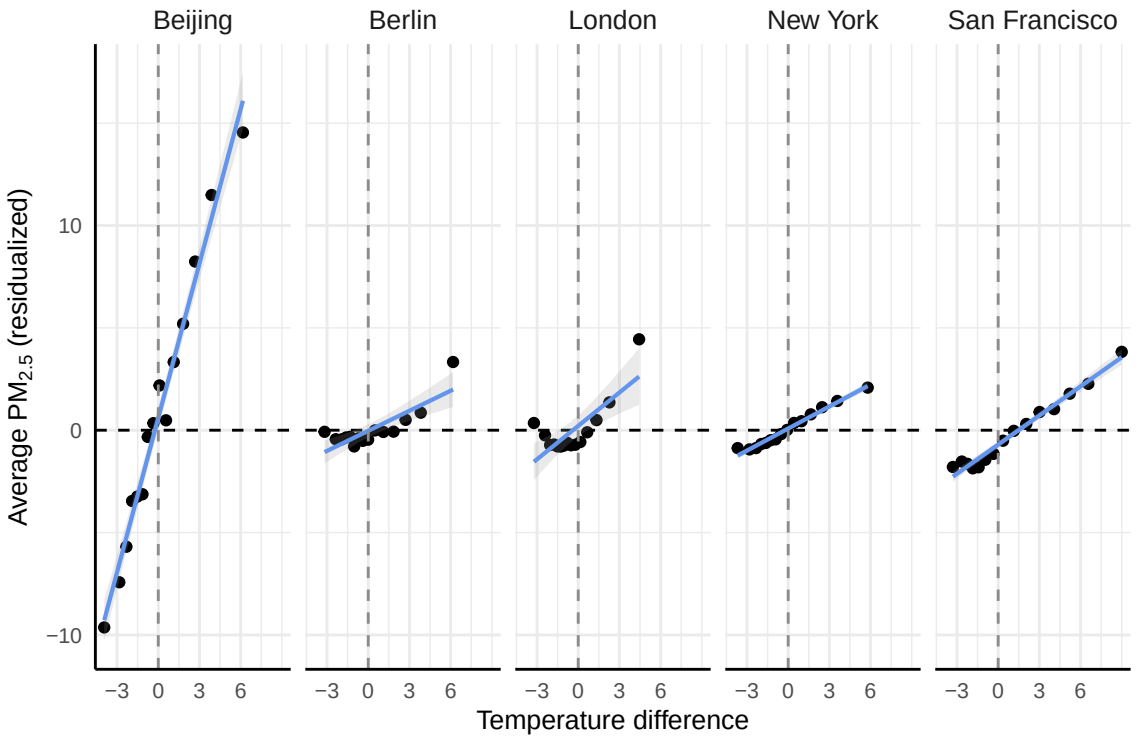


Figure 5: Illustration of the First Stage

Notes: Binned scatter plots of residualized $PM_{2.5}$ and inversion strength in five city groups. Blue lines: linear fit. Shaded areas: 95% Confidence Intervals. Groups: 15 East Asian cities (*Beijing*), 21 Central and Northern European cities (*Berlin*), 35 cities across the UK, Ireland and Northern France (*London*), 21 cities in the Eastern US and Canada (*New York*), 30 cities in and around Silicon Valley (*San Francisco*).

of-week cell. This measure captures non-linear effects of pollution and allows these to vary by location and time. 10.7% of all city × day observations are characterized by such a $PM_{2.5}$ shock.

5 Main Results

In this section, we first present results on how $PM_{2.5}$ exposure affects the quantity and quality of output developers produce. Thereafter we show that they use two margins of adjustment, task choice and working hours, to adapt to increases in pollution concentration.

5.1 Work Quantity and Quality

Table 2 displays 2SLS estimates of the effect of $PM_{2.5}$ exposure on output quantity and quality. As we test ten hypotheses in total, we report p-values corrected for multiple hypothesis testing following the Benjamini-Hochberg procedure in squared brackets. We find that an increase by 1 $\mu g/m^3$ causes developers' overall output, measured by total actions, to fall by 0.0021 or 0.08% of the sample mean. To explore the contribution of the extensive and intensive margins, respectively, to the overall reduction in work quantity, we use an indicator for a positive activity

Table 2: IV Estimates of the Effect of $PM_{2.5}$ on Work Quantity and Quality

	Actions (1)	*Any actions* (2)	*PR acceptance rate* (3)	*Commits* (4)	*Comments* (5)
Panel A: Levels					
$PM_{2.5}$	-0.0021***	-0.0002***	-0.0004***	-0.0009***	-0.0007**
	(0.0006)	(0.00005)	(0.0001)	(0.0003)	(0.0003)
	[0.0017]	[0.0019]	[0.0017]	[0.0017]	[0.0428]
First stage F-stat.	171.3	171.3	199.3	171.3	171.3
% change	-0.083	-0.048	-0.063	-0.079	-0.078
% of full effect		19.8			
Panel B: Shocks					
$PM_{2.5}$ shock	-0.1084**	-0.0080**	-0.0131*	-0.0604***	-0.0311
	(0.0546)	(0.0040)	(0.0067)	(0.0227)	(0.0256)
	[0.0572]	[0.0572]	[0.0572]	[0.0159]	[0.2251]
First stage F-stat.	110.3	110.3	87.2	110.3	110.3
% change	-4.30	-2.31	-1.98	-5.12	-3.68
% of full effect		18.6			
Observations	856,522	856,522	334,060	856,522	856,522
Dependent variable mean	2.52	0.345	0.663	1.18	0.846

Notes: Regressor of interest is $PM_{2.5}$ in $\mu g/m^3$ in Panel A, or a dummy as in Equation (3) in Panel B. Covariates: Eight bins for mean daily temperature, third-order polynomials in wind speed, precipitation, and relative humidity, indicators for heavy wildfire smoke and holidays, as well as city, region-by-day-of-week, country-by-month, and region-by-year-by-month fixed effects. Coefficients on weather controls can vary across regions. Regressions weighted by number of developers in city-month cell. Standard errors, clustered by city, in parentheses. Benjamini-Hochberg p-values in squared brackets. Significance levels (based on conventional p-values): $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

level in column 2. We find that the extensive margin effect contributes approximately 20% to the full reduction in actions. Given that our sample of GitHub users likely comprises mostly young and middle-aged adults, they are unlikely to suffer severe health damages from short-run pollution exposure which prevent them from working, but rather experience subtle effects on health and cognitive function, reducing their productivity. Output quality, as measured by the the PR acceptance rate, also drops by 0.4 percentage points or 0.06% of the mean acceptance probability. For an increase in $PM_{2.5}$ by one within-city standard deviation ($16.4\,\mu g/m^3$), the estimates imply reductions of 1.4% and 1.0%, respectively, in the quantity and quality measures. In columns 4 and 5 we investigate what type of activities is driving the reduction in output quantity. Both the number of commits, the primary measure of individual coding activity, and the number of comments written in discussion fora, as a measure of collaborative work, drop by 0.08% relative to their sample means, indicating that the pollution-induced productivity shock reduces productivity in both types of work.

To investigate whether the effect of pollution exposure is linear, or rather driven by unususally high daily concentrations, we use the binary $PM_{2.5}$ shock variable as regressor in panel B. The 2SLS estimates imply that on a day with unusually high pollution, the number of total actions falls by 0.108 or 4.3% of the mean. This is mostly driven by core coding activity, as the number of commits falls by 5.1%, while the effect on comments is smaller and not statistically significant. The contribution of the extensive margin is again close to 20%. In

addition, output quality drops by 2% relative to the sample mean. Across specifications, the first stage F-statistics on the excluded instruments are high, indicating that the IVs are sufficiently strong.

In Appendix Table A.6, we investigate less frequent action types—the number of issues and PRs opened and closed, respectively. Like a commit, opening a PR reflects individual coding work. Closing a PR implies reviewing and deciding whether to accept or reject proposed code changes, and opening/closing issues generally starts/ends a discussion with other users. The strongest negative effects arise for the number of new issues opened, which drops by 0.19% and 8% relative to the mean, in response to a unit increase in $PM_{2.5}$ and a $PM_{2.5}$ shock, respectively. For the other action types, we find significant reductions only for $PM_{2.5}$ in levels, of a similar relative magnitude as for commits and comments.

In sum, these results imply that fine particulate matter exposure exerts a negative effect on both the quantity and quality of developer output, which are strongly driven by substantial declines on days with relatively poor air quality. While the drop in quantity is uniform across action types for $PM_{2.5}$ in levels, developers mostly reduce individual coding activity and work on new tasks (commits, opening new issues), when pollution reaches unusually high levels, whereas effects are smaller for activities that respond to work conducted by others (writing comments, closing issues and PRs). This suggests that workers in collaborative work environments adapt work patterns and prioritize more interactive activities when hit by an adverse productivity shock.

Columns 1 to 5 of Table A.7 show OLS estimates for the key quantity and quality measures. We obtain negative estimates for all outcomes with $PM_{2.5}$ in levels, but they are significant only for the quantity measures. With the $PM_{2.5}$ shock, coefficients are insignificant. Mirroring a common finding in the literature, almost all estimates are substantially smaller in absolute magnitude than the 2SLS results, pointing towards attenuation bias due to measurement error.

Effect Dynamics. To explore effect dynamics, we use a "reduced form" version of our model, regressing output quantity and quality on four variables measuring inversion strength in city c on day d and each of the previous three days. Appendix Table A.8 shows that for total actions, commits, and the indicator for any actions, only same-day inversion strength generates negative and significant effects, whereas the point estimates on the lags are close to zero and statistically not significant. Only the PR acceptance rate is negatively affected by both same-day and lagged inversion strength, with significant coefficients on the second and third lag. Thus, in our sample, adverse effects of acute pollution exposure arise immediately and are mostly limited to the same day, with potentially slight lagged effects on cognitive performance.

Effect Magnitude. Is the effect of $PM_{2.5}$ on developers' output economically meaningful? To assess this, we first benchmark it against the impact of another highly relevant environmental shock, exposure to extreme outdoor temperatures, which have been found to affect, e.g., student performance, social media sentiment, and mental health (Park, 2020; Baylis, 2020;

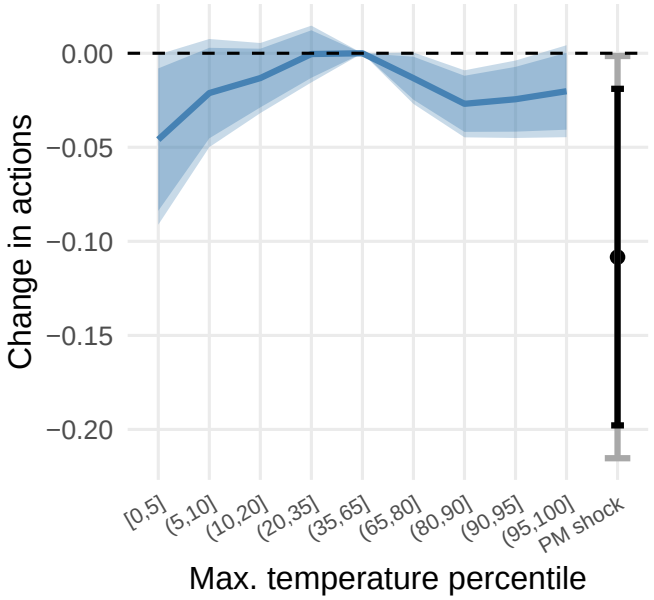


Figure 6: Effects of $PM_{2.5}$ and heat on work quantity

Notes: Black point and error bands: IV estimate with 90 and 95% confidence intervals for $PM_{2.5}$ shock from Table 2. Blue lines and shades: OLS estimates with 90 and 95% confidence intervals for maximum daily temperature range indicators shown on the x-axis. Covariates: Flexible controls for minimum temperature, precipitation, wind speed and relative humidity, indicators for heavy wildfire smoke and holidays, as well as city, country-by-month, region-by-day-of-week, and region-by-year-by-month fixed effects. Standard errors clustered by city. Regressions weighted by number of developers in city-month cell.

Mullins and White, 2019). Figure 6 reproduces the estimated effect of a $PM_{2.5}$ shock on total actions in graphical form (point estimates with confidence intervals displayed in black) along with coefficients from OLS regressions of total actions on maximum daily temperature, measured with eight dummy variables indicating different percentile ranges of the city-specific distribution. The reference category is maximum temperature between the 35th and the 65th percentile. The effects of temperature follow an inverse u-shape: Both unusually cold and unusually hot temperatures have adverse effects. Even though software developers might work in climate-controlled offices, heat and cold exposure during commutes or while running errands could generate these negative effects. While the IV estimate for the $PM_{2.5}$ shock, which occurs on 10% of all days, is less precise than the OLS estimates for temperature, the point estimate is more than twice the estimates for the temperature bins representing extreme cold and extreme hot temperatures. Hence, knowledge worker productivity appears to be substantially more vulnerable to poor air quality than to extreme temperatures, an environmental shock of high relevance given global warming.

Secondly, we use the average monetary value of commits and PRs (derived in Section 3.3) to translate the estimated effects of $PM_{2.5}$ exposure into changes in output value.[11] For a $1\,\mu g/m^3$ increase in $PM_{2.5}$ the implied reduction in output value amounts to \$0.147 per software

[11]We assume a value of \$112 per commit (Section 3.3). For PRs, we do not use the mean value of \$354 found in the Gitcoin data because PRs in that sample are on average larger than PRs created in our sample. Instead, we value PRs with $2.7 \times \$112 = \$300$ given that they comprise, on average, 2.7 commits. We use these numbers to value changes in commits, PRs opened and the PR acceptance rate.

developer and day. A permanent increase would consequently reduce annual output value by \$53.66 per developer. For comparison, Deryugina et al. (2019) estimate annual mortality and medical cost of a permanent 1 $\mu g/m^3$ increase in $PM_{2.5}$ of \$117 per medicare beneficiary. On days with a $PM_{2.5}$ shock, output value falls by \$7.3 relative to days with better air quality. Eliminating pollution spikes, relative to the location- and season-specific averages, would generate annual benefits of as much as \$266 per worker.[12] As we ignore other losses, for example, from less work on issues, these estimates can be interpreted as lower bounds. Extrapolating to all US workers in the occupation group 'Computer and Mathematical Workers' and all ICT professionals in the European Union suggests annual benefits of \$250m (US) and \$420m (EU), respectively from permanently reducing $PM_{2.5}$ by 1 $\mu g/m^3$.[13] Eliminating $PM_{2.5}$ shocks, relative to the city × month × day-of-week specific average, could generate annual benefits of as much as \$1.24b (US) and \$2.09b (EU).

The results so far indicate, that acute exposure to air pollution causes quantitatively meaningful reductions in knowledge workers' output. They also suggest that workers respond, in particular to unusually high pollution, by prioritizing certain tasks, especially more interactive ones. In the next section, we investigate worker adjustment in more detail.

5.2 Worker Adjustment

To investigate whether work patterns change in response to increases in air pollution we consider two potential margins of adjustment: task choice and working hours.

Task Choice. We start by analyzing whether developers switch to easier tasks when exposed to higher levels of pollution. To assess the complexity of activities related to pull requests, i.e., coding and code review tasks, we consider three PR characteristics: lines of code added, lines of code deleted, and number of files changed, averaged across all PR activities on a given day.[14] While there can be very difficult tasks that require only small changes in the code, we believe that these variables provide reasonable proxies of PR complexity. Fixing a severe bug likely requires changes in various parts of the code, implying a larger number of files changed. Results in Table C.3 indicate that PRs with more lines of code added and files changed are rewarded higher payments on Gitcoin, validating the use of these variables as complexity metrics. Similarly, reviewing a PR is clearly more demanding when it contains large changes across different files.

[12] This number is computed as \$7.3 per day × 36.5 days, as pollution shocks occur on 10% of all days.

[13] Given 4,654,750 'Computer and Mathematical Workers' in the US in 2021 (Bureau of Labor Statistics, 2022), the annual gain in output value of a permanent one $\mu g/m^3$ pollution reduction for this group is computed as \$0.147 per day × 365 days × 4,654,750 = \$250m. We compute the value for the EU analogously, based on 7,843,000 ICT professionals in 2020 (Cedefop, 2022).

[14] These variables are based on GHArchive data, while work quantity results use GHTorrent data. GHArchive data are only available from 2015 onward. Appendix Table A.9 shows that using data on PRs from GHArchive and GHTorrent yields very similar results for the effect of $PM_{2.5}$ on the number of PRs opened or closed.

Table 3: IV Estimates of the Effect of $PM_{2.5}$ on Task Complexity

	Lines added per PR (1)	*Lines deleted per PR* (2)	*Files changed per PR* (3)	*PR complexity* (4)	*Share easy issue events* (5)
Panel A: Levels					
$PM_{2.5}$	-0.0011**	-0.0010**	-0.0008***	-0.0006***	0.00002
	(0.0005)	(0.0004)	(0.0002)	(0.0002)	(0.00003)
First stage F-stat.	192.8	192.8	192.8	192.8	167.3
% change	-0.108	-0.097	-0.077		0.067
Panel B: Shocks					
$PM_{2.5}$ shock	-0.0416*	-0.0267	-0.0335***	-0.0216**	-0.0004
	(0.0228)	(0.0261)	(0.0121)	(0.0098)	(0.0011)
First stage F-stat.	94.6	94.6	94.6	94.6	109.0
% change	-4.08	-2.64	-3.30		-1.29
Observations	374,588	374,588	374,588	374,588	562,922

Notes: Panel A: Regressor is $PM_{2.5}$ in $\mu g/m^3$. Panel B: Regressor is a dummy as in Equation (3). In Columns (1) to (3), outcomes are transformed by the inverse hyperbolic sine. Covariates: See Section 4 or Table 2. Regressions weighted by number of developers in city-month cell. Standard errors, clustered by city, in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

We apply the inverse hyperbolic sine transformation to the three variables such that coefficients approximate percentage changes. Table 3 presents the results, showing that on top of the overall reduction in the number of actions completed, developers switch towards less complex coding and review tasks in response to $PM_{2.5}$-induced productivity declines. The sample size is reduced as the outcomes are defined only for city × day observations with any PRs opened or closed. An increase in $PM_{2.5}$ by 1 $\mu g/m^3$ reduces all three measures of PR complexity, with effect sizes between -0.08% and -0.11% (Panel A). On days with a $PM_{2.5}$ shock, the number of files changed and of lines added fall by 3.35% and 4.16%, respectively, indicating that developers respond mostly to large productivity shocks on high-pollution days (Panel B). The effect on the number of lines deleted is also negative, but not significant. This pattern is consistent with the idea that tasks related to deleting code, e.g., cleaning a file or dropping a redundant part, are usually easier to do and to review than tasks related to creating new code. Column (4) shows results for a standardized PR complexity index that combines all three aforementioned variables.

In Column (5), we analyze whether developers also focus on easier issue-related tasks, using the share of issue events completed that refer to an easy issue as outcome. For this dimension of task complexity, we do not find any significant or sizeable effects of $PM_{2.5}$ exposure. A potential explanation might be that work related to issues (commenting, closing discussions), might be less complex overall than coding and code review work.

Heterogeneity by Experience. Next, we investigate whether the adjustment strategy of focusing on easier tasks is used equally by all software developers. Specifically, we separately estimate the effects of pollution on output and task choice across subsamples based on experi-

Table 4: Effect Heterogeneity in Pollution Effects on Adjustment and Output by Experience

	Actions	*PR acceptance rate*	*PR complexity*	*Share easy issue events*
	(1)	(2)	(3)	(4)
Panel A: Bottom experience tercile				
$PM_{2.5}$	-0.0012**	-0.0004**	-0.0006**	0.00007***
	(0.0006)	(0.0002)	(0.0003)	(0.00003)
Dependent variable mean	2.24	0.689		0.032
% change	-0.055	-0.052		0.217
First stage F-stat.	192.2	227.3	189.0	163.8
Observations	6,517,497	426,407	596,762	1,151,662
Panel B: Mid experience tercile				
$PM_{2.5}$	-0.0028**	-0.0007***	-0.0003	0.00004
	(0.0014)	(0.0002)	(0.0003)	(0.00005)
Dependent variable mean	2.44	0.648		0.029
% change	-0.115	-0.104		0.149
First stage F-stat.	125.8	156.8	144.6	114.8
Observations	7,024,992	486,915	683,597	1,440,602
Panel C: Top experience tercile				
$PM_{2.5}$	-0.0033**	-0.0002	0.00004	-0.00010*
	(0.0014)	(0.0003)	(0.0004)	(0.00006)
Dependent variable mean	2.92	0.658		0.027
% change	-0.113	-0.032		-0.349
First stage F-stat.	131.6	141.6	148.5	148.1
Observations	6,033,820	467,195	706,849	1,421,717

Notes: Regressions are estimated at the developer × date level using two-stage least squares. The samples used in Panels A to C are based on developer experience on GitHub which is measured by an index that combines time since registration on GitHub and the number of followers a developer had at the time they first enter the estimation sample. Covariates: developer, region-by-day-of-week, country-by-month, and region-by-year-by-month fixed effects, cubic polynomial in developers' time since registration on GitHub as well as weather covariates as described in Section 4. Standard errors, clustered by city, in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

ence. Especially inexperienced workers might have a large incentive to switch to easier and smaller tasks, as (i) they might be most prone to commit errors when hit by a productivity shock, and (ii) the *quantity* of daily contributions is visualized prominently on GitHub user profiles, whereas their size or complexity is harder to assess. As potential employers are known to check GitHub profiles of candidates (Gupta et al., 2024), especially junior developers might adjust task choice to keep up high activity levels when hit by a productivity shock. We build an experience index that combines the developer's time since registration on GitHub and number of followers at the point that he enters the sample. We then repeat the estimations of the effect of $PM_{2.5}$ concentrations on the key output quantity and quality measures as well as the measures of task choice (PR complexity index and share easy issue events), separately for developer groups in the different terciles of the experience index (Table 4). Developers in the bottom tercile on average have 3.1 years of experience on GitHub and 21 followers,

whereas developers in the upper terciles have more than 6 years of experience and 85 followers on average. Indeed, when comparing developers in the bottom and middle tercile, we find that inexperienced developers respond more to pollution exposure on both dimensions of task complexity, shifting to less complex PRs and easier issues. Conversely, they see a smaller impact on output quantity and quality, both in relative and absolute terms. These patterns underscore the role of switching to easier and smaller tasks as an adjustment response among knowledge workers, and the benefits of worker flexibility in terms of determining tasks and priorities for alleviating the adverse impacts of productivity shocks on output. The most experienced developers do not shift towards easier tasks in response to pollution, but even tackle fewer easy issues as $PM_{2.5}$ increases. This pattern is consistent with a congestion mechanism: when less experienced developers disproportionately claim easy tasks, the stock of such tasks available to experts shrinks, leaving them to take on more difficult work. Consequently, the most experienced developers exhibit the largest decline in the total number of actions. Output quality, however, does not change significantly in response to an increase in $PM_{2.5}$, suggesting that extensive experience to some extent insulates quality from this shock.

Overall, the average effects mask important heterogeneity where junior developers respond more to a pollution-induced productivity shock by reallocating effort to easy tasks, while senior developers have to concentrate on the complex issues and therefore see the largest reduction in output quantity.

Working Hours. In many jobs, knowledge workers are flexible in when and where they want to work. Thus, a second adjustment margin available is a change in working hours. Developers might for example work longer hours in the evening to compensate for the productivity decline to reach a certain target for the day, or conversely, reallocate work activity from high-pollution, low-productivity to low-pollution, high-productivity days. Table 5 presents the estimated effects of daily $PM_{2.5}$ on same-day working hours, measured by the timestamp of the first and the last action, respectively, performed by a developer on a given day (in minutes).

With the linear $PM_{2.5}$ measure, we find very limited effects on working hours. The significant coefficient on the timestamp of the first action indicates that for a $10\,\mu g/m^3$ increase in $PM_{2.5}$, developers on average complete the first action of the day 1.28 minutes later. A $PM_{2.5}$ shock, by contrast, induces a stronger response, causing developers to end the work day on average 5 minutes earlier than on days with better air quality, with no significant effect on the start of the work day. Consequently, the average developer is not using the evening hours to compensate for the adverse productivity shock. Instead, when a pollution shock makes them feel unproductive, they decide to end work activity earlier, likely due to the low returns to work effort.

Do developers make up for the earlier end of the work day by reallocating work time to lower pollution, and thus higher-productivity, days? To investigate this, we estimate the effect of $PM_{2.5}$ shocks during the workweek on output produced on the following weekend.

Table 5: Effect of $PM_{2.5}$ on Working Time (IV Estimates)

	Time of first action (minutes)		*Time of last action (minutes)*	
	(1)	(2)	(3)	(4)
$PM_{2.5}$	0.128**		-0.022	
	(0.056)		(0.043)	
$PM_{2.5}$ shock		-2.11		-5.12**
		(2.35)		(2.31)
First stage F-stat.	189.8	139.6	189.8	139.6
Observations	675,304	675,304	675,304	675,304

Notes: Panel A: Regressor is $PM_{2.5}$ in $\mu g/m^3$. Panel B: Regressor is a dummy as in Equation (3). Dependent variables are the time of day of the first and last action a developer performed that day, respectively, measured in minutes since midnight. Covariates: See Section 4 or Table 2. Regressions weighted by number of developers in city-month cell. Standard errors, clustered by city, in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Since activity levels on weekends are on average positive, but low compared to working days, weekends offer most scope for compensation (see Figure 2). This analysis is conducted at the developer × week level, using the following regression model:

$$y_{i,c,w}^{\text{weekend}} = \beta PM_{c,w}^{\text{Mo–Fr}} + \mu_i + g(tenure_{i,c,w}) + \mathbf{w}'^{\text{weekend}}_{c,w}\boldsymbol{\gamma_{R(c)}} + \mathbf{w}'^{\text{Mo–Fr}}_{c,w}\boldsymbol{\alpha}_{R(c)} + \delta_{R(c)}h_{c,w} + \mu_{R(c),yr(w),q(w)} + \mathbf{z}'^{\text{weekend}}_{c,w}\boldsymbol{\varphi} + \varepsilon_{i,c,w}. \quad (4)$$

Here, $y_{i,c,w}^{\text{weekend}}$ denotes work time or output of developer i living in city c on the weekend of week w. To measure work-week $PM_{2.5}$ exposure, $PM_{c,w}^{\text{Mo–Fr}}$, we either use the number of days with a $PM_{2.5}$ shock between Monday and Friday, or an indicator for at least one $PM_{2.5}$ shock during this period. Since exposure to pollution reduces output quantity only on the same day, but not the following days (Appendix Table A.8), the coefficient β should pick up developers' *behavioral* adjustment to a productivity shock during the workweek, and not confound it with *physiological* effects.

Pollution is instrumented by average nighttime inversion strength between Monday and Friday, interacted with indicators for the first-stage city groups g. To account for auto-correlation in the instruments, we add the vector $\mathbf{z}_{c,w}^{\text{weekend}}$ to the model, which includes the instrumental variables measured on the weekend. This ensures that we do not pick up the effects of inversion-induced changes in pollution on the weekend itself. The model further includes a developer fixed effect μ_i, a cubic polynomial in the developer's tenure on GitHub, $g(tenure_{i,c,w})$, two sets of weather controls, $\mathbf{w}_{c,w}^{\text{Mo–Fr}}$ and $\mathbf{w}_{c,w}^{\text{weekend}}$, covering the exposure period and the weekend, respectively,[15] the number of public holidays during the workweek, $h_{c,w}$, and a region-by-year-by-quarter fixed effect, $\mu_{R(c),yr(w),q(w)}$. In models with working time variables as outcomes, we further include region-by-calendar week fixed effects to improve precision, as

[15]The vectors $\mathbf{w}_{c,w}^{Mo-Fr}$ and $\mathbf{w}_{c,w}^{\text{weekend}}$ contain the number of days with wildfire smoke exposure, third-order polynomials of average temperature and quadratic polynomials of average precipitation, relative humidity, and wind speed during the respective period.

Table 6: Effect of $PM_{2.5}$ on Weekend Work (IV Estimates)

	Time of first action		*Time of last action*		*PR complexity*		*Actions*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Days with $PM_{2.5}$ shock	-4.189*		-0.7019		0.0215**		0.0636	
	(2.365)		(1.881)		(0.0091)		(0.0489)	
Any $PM_{2.5}$ shock		-9.583*		-1.302		0.0489**		0.1175
		(5.795)		(4.476)		(0.0213)		(0.1112)
First stage F-stat.	59.3	64.1	59.3	64.1	47.5	69.6	55.3	65.0
Observations	563,297	563,297	563,297	563,297	139,881	139,881	1,826,522	1,826,522

Notes: 2SLS regressions are estimated at the developer × week level. The regressor of interest is either the number of days with a PM2.5 shock between Monday and Friday, or an indicator for at least one $PM_{2.5}$ shock between Monday and Friday. Models are estimated only on weeks with PM2.5 below the city-specific 75th percentile on both weekend days. Covariates: see Equation 4. Standard errors, clustered by city, in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

work time on the weekend likely responds to seasonal changes in local sunrise time. To analyze whether developers reallocate work from high-pollution work days to low-pollution, high productivity weekends, – in line with the intertemporal substitution hypothesis derived from a standard labor supply model – we estimate the model only on weekends without unusually high air pollution levels, defined as levels below the city-specific 75th percentile on both days.

Results in Table 6 indicate that developers indeed devote more time to work on GitHub on the weekend if they were exposed to pollution-induced productivity shocks during the work week. They start working significantly earlier, by 4.2 minutes for every day with a $PM_{2.5}$ shock during the work week, or by 9.6 minutes on average after weeks with at least one shock. The time of the last action, by contrast, is unaffected, implying a longer overall time input. On average, starting work earlier on low-pollution weekends compensates for 82% of the reduction in work time due to the earlier end of the work day (5.12 minutes). The results further indicate that developers use the extra time they invest after work weeks with unusually high pollution to address more complex tasks, compensating for the shift towards easier tasks on work days with $PM_{2.5}$ shocks: The complexity of PRs worked on increases by 2.15% of a standard deviation on average for every shock experienced on work days, fully compensating for the same-day adverse effect (-2.16%). Developers also tend to produce more output overall on weekends if they were exposed to unusually high levels of $PM_{2.5}$ during the workweek, as indicated by the positive point estimates for total actions, but the effects are not significant.[16] This is consistent with the fact that developers tackle more complex, and likely more time-consuming tasks. It also implies that, while knowledge workers do adjust to mitigate the negative impact of a productivity shock, air pollution still causes a net reduction in overall output and thus meaningful economic costs.[17]

To check that the estimates in Table 6 indeed reflect a behavioral response of developers to pollution-induced productivity shocks, we conduct a falsification test. We shift both the

[16] The insignificant point estimate is equivalent to 59% of the reduction in actions on days with a $PM_{2.5}$ shock.
[17] We do not find any reallocation of work activity to weekends with high pollution levels (Appendix Table A.10).

weekend and the exposure period forward by two days. The placebo weekend comprises Monday and Tuesday and the placebo exposure period ranges from Wednesday to Sunday of the week before. Since activity levels are low on weekends, productivity shocks during the placebo exposure period, which partially falls on the weekend, should not induce strong compensation during the following week. Moreover, activity is already high on Monday and Tuesday, such that there is not much scope for additional work. Hence, we expect no significant effects of $PM_{2.5}$ exposure. Appendix Table A.11 presents the results which confirm this hypothesis.

The flexibility to shift work activity to low-pollution weekends allows developers to end work early on high-pollution work days, which might mitigate adverse effects of $PM_{2.5}$ on performance: If developers can quit working when their health or cognitive capacity deteriorates and they face an increased risk of committing errors, impacts of pollution on same-day work quality are alleviated. At the same time, sacrificing leisure on the weekend, when it is likely most valuable, implies a welfare cost and potentially adverse effects on the work-life balance.

6 Heterogeneity and Further Results

The main results show that even in a highly flexible knowledge work setting, where workers can adjust work time and task choice in response to productivity shocks, air pollution significantly reduces output quantity and quality, and these effects translate into relevant economic costs. In this section, we explore potential mechanisms and investigate whether these damages are equally distributed across locations, exploiting the broad geographic variation in our sample.

Non-Linearity. We start by investigating whether the effect of pollution on output varies across the range of $PM_{2.5}$ concentrations. To analyze the shape of the dose-response function, we replace $PM_{c,d}$ in equation (1) with a series of indicator variables for $PM_{2.5}$ concentration falling into a specific bin.[18] Since we do not have enough instruments for running 2SLS, we opt for a more conservative specification with stricter fixed effects for region × date and city × month. These absorb (i) region-wide shocks to developer output on a given date that might be correlated with $PM_{2.5}$ concentration and (ii) seasonal fluctuations in activity and air quality which are allowed to vary across cities.[19]

Figure 7 displays the estimated impact on total actions when moving from a $PM_{2.5}$ concentration below 4 $\mu g/m^3$ to the respective bin. We find a significant reduction in the number of daily actions starting already at levels below current limit values in place, e.g., in the US and Europe. The slope of the dose-response function is steepest at low concentrations, and becomes somewhat flatter at higher pollution levels. The results imply that even in cities with

[18] Bins are ≤ 4, (4-8], (8-16], (16-20], (20,25], (25,30], (30,40], (40,50], (50, 70], (70, 100], (100, 200], and > 200 $\mu g/m^3$.

[19] OLS results from this model with either $PM_{2.5}$ in $\mu g/m^3$, or the indicator for a $PM_{2.5}$ shock as regressors are presented Columns 4 to 6 of Table A.7.

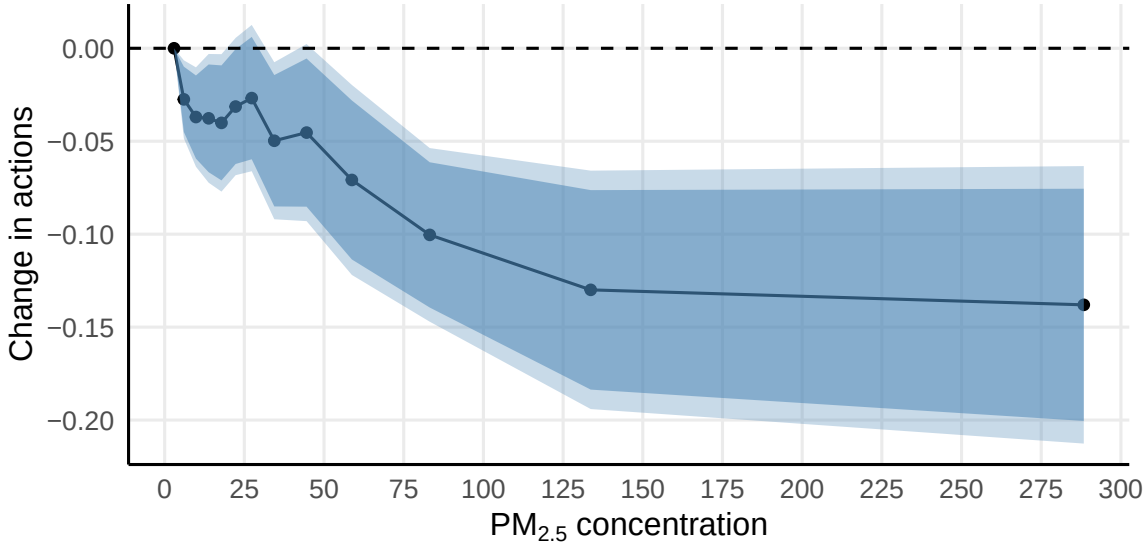


Figure 7: Non-Linear Effects of $PM_{2.5}$ on Output Quantity

Notes: OLS estimates from regression of total actions on different bins of $PM_{2.5}$ concentrations. Covariates: Weather and holiday controls as in Section 4, region-by-date and city-by-month fixed effects. X-axis: Average $PM_{2.5}$ concentration in each bin in $\mu g/m^3$. Shaded areas: 95% and 90% confidence intervals.

low to moderate levels of $PM_{2.5}$, further improvements in air quality will generate positive effects on worker productivity.

In Appendix Figure B.5 we zoom in on the different parts of the function, by grouping the cities into terciles based on average pollution concentration. The most pronounced negative effects of increased pollution on total actions occur in the bottom and middle terciles at $PM_{2.5}$ concentrations below 15 $\mu g/m^3$. Given that the OLS estimates likely underestimate the true effects of $PM_{2.5}$ exposure, the results imply relevant economic benefits from complying with the WHO guideline for $PM_{2.5}$. Notably, at the largest absolute concentrations in the two tercile groups, the dose-response functions become flat, suggesting that developers might become aware of pollution when it reaches "extreme" levels relative to what is normal in their city of residence, and start adapting. We find no significant effects of increases in $PM_{2.5}$ in the upper tercile. This could be attributable to individuals in cities with higher pollution being more aware of air pollution's adverse effects, and engaging more in avoidance behavior, like using air purifiers, which improve indoor air quality.[20]

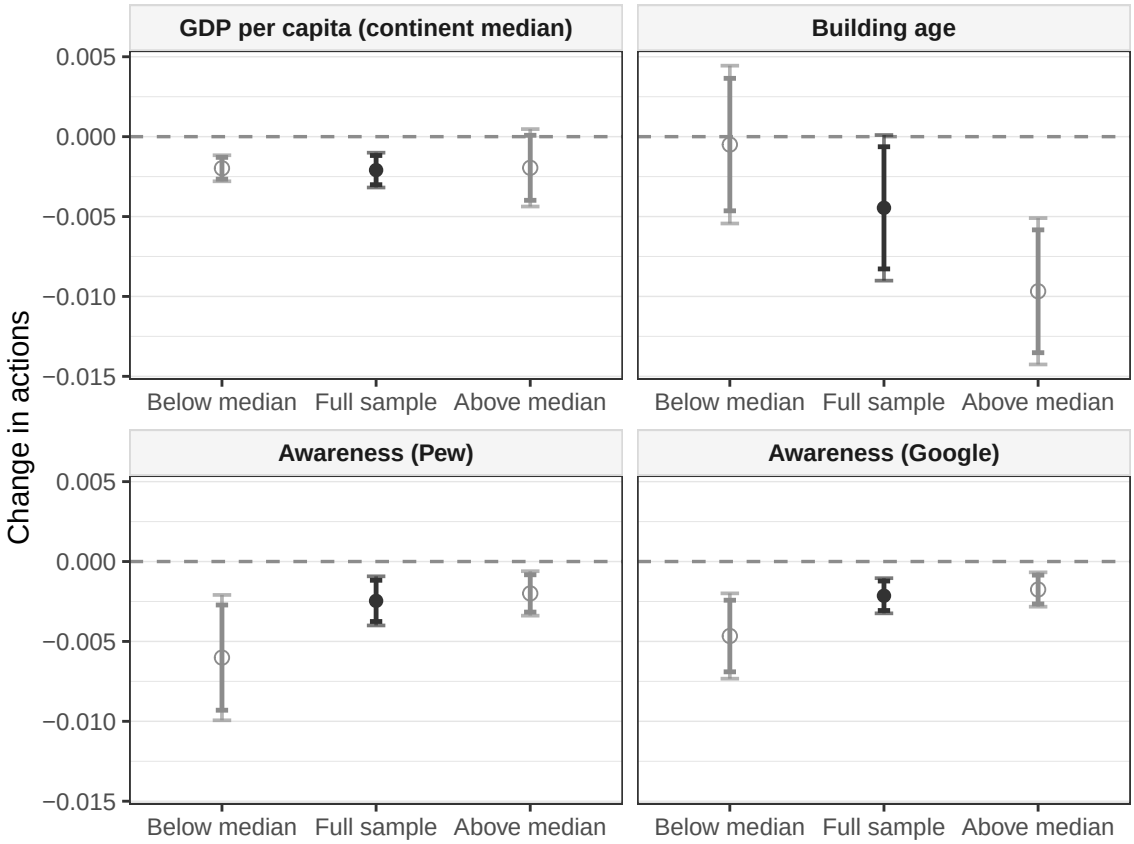
Effect Heterogeneity. Next, we analyze effect heterogeneity by three location characteristics, local income, building stock age, and air pollution awareness (Figure 8, Table A.12).

For income, we group cities based on whether their GDP per capita in 2014 was above or below the *continent-specific median*.[21] This approach alleviates concerns that we confound

[20]There is anecdotal evidence that big (tech) companies equip their offices in highly polluted places with air purifiers and filters, e.g., Microsoft, Google, SAP and Coca-Cola in Delhi or Nokia in Beijing.

[21]The main source for local GDP per capita is the OECD metropolitan area database. We assign to each city the value for its respective metro area. Data for cities in non-OECD countries come from national statistical agencies, the OECD regional statistics database, or the World Bank. All values are adjusted for purchasing power parity.

Figure 8: Effect Heterogeneity by Local Income, Building Stock and Awareness (IV Estimates)



Notes: All markers depict 2SLS estimates of the effect of a 1 $\mu g/m^3$ increase in $PM_{2.5}$ concentration on the number of total daily actions, as in Table 2 (Column 1, Panel A), along with 95% and 90%-confidence intervals. The four subplots depict results form heterogeneity analyses based on city-level GDP per capita (in 2014), local building stock age (share of dwellings built before 1960), and air pollution awareness (measured either by the share of people stating to view air pollution as a big problem in the 2020 Pew Study, or by google search interest for air pollution related terms (in 2014). The black markers depict estimates based on all city × date observations for which data on the respective variable is available. The grey markers depict estimates from the subsamples with above and below median values for the variable in question, respectively. Detailed results are depicted in Table A.12.

local income and other regional differences correlating with income. We find almost identical absolute effects of a unit increase in $PM_{2.5}$ concentration on total output across subsamples. Relative to the sample mean, effects are slightly larger in the low-GDP sample (0.09% vs 0.07%). These results deviate from the common finding that lower-income areas tend to suffer more harm from similar environmental hazards (e.g. Colmer et al., 2021; Hsiang et al., 2019). A potential explanation might be that software developers are highly-skilled and highly paid workers, i.e. even in poorer locations, the individuals in our sample are relatively affluent.

Effective exposure to particulate matter is likely lower for individuals in modern buildings with low penetration rates than for those in older, lower-quality buildings given the same outdoor concentration. We use data on the construction period of residential dwellings as a proxy for building stock quality, covering 322 sample cities.[22] We categorize cities into

[22]The data are collected from the American Community Survey for metropolitan areas, the Federal Statistical Offices of Germany and Switzerland (local level), the EU Building Stock Observatory (country-level), the Statistics Bureau of Japan (prefecture-level), Statistics Canada (province-level), and Statistics Norway (municipality-level).

two groups based on the share of dwellings built before 1960. We find pronounced effect heterogeneity by building stock age: The negative effect of $PM_{2.5}$ on total actions is strongly driven by the sample with an above median share of old dwellings (50% of buildings built before 1960, on average), whereas the point estimate is close to zero for cities with a relatively modern building stock (9% of buildings built before 1960, on average). In the subsample of cities with a relatively old building stock, a 10 $\mu g/m^3$ increase in PM2.5 reduces total daily actions by 3.6% of the sample average. The finding that developers in locations with low building stock quality, and thus likely higher indoor exposure to $PM_{2.5}$, bear a disproportionate share of the overall effect, suggests that the main results are attributable to physiological responses to air pollution, rather than behavioral changes.

To assess the potential role of avoidance behavior, we investigate how the effect of $PM_{2.5}$ differs between regions with low and high awareness of air pollution as a significant issue. We use data from the 2020 Pew Research Center International Science Survey to derive a country-wide measure of awareness, namely the share of respondents stating that air pollution is a big problem in their country. 167 of our sample cities in 18 countries are covered by the survey data and we split them based on median awareness.[23] As an alternative, we use data from google trends on search interest by country for terms related to air pollution (*PM2.5, PM10, air pollution*) in 2014, and assign cities into the high awareness group if they have above median search interest for pollution related terms.[24] Even though they cover different samples, we find consistent results with the two awareness measures: While effects of $PM_{2.5}$ on total actions are negative and significant in both subsamples, they are two to three times larger in the low awareness countries as compared to the locations with high awareness. The pattern suggests that the reduction in output is not generated by avoidance behavior, e.g., working from home or going on vacations on high-pollution days. In such a case we would expect to find increasing effect magnitude in awareness. Rather, awareness seems to alleviate effects. This may be due to higher ownership and utilization of protective devices like air purifiers or masks in the high awareness group.

Overall, we find that knowledge worker vulnerability to air pollution does not differ systematically based on local income levels, but instead varies with awareness of and physical exposure to the environmental hazard. Policy interventions related to modernizing building infrastructure and raising awareness for adverse health impacts and local levels of air pollution, might thus generate positive effects on the local economy.

Robustness. We show that our main results for work quantity, quality, and task choice are robust to changes in the specifications of the first and second stage models, respectively. In the first stage model, we (i) replace inversion strength, $\Delta T_{c,d}$, by an inversion indicator,

[23] Cities in the above (below) median sample on average have a share of 69% (53%) of respondents believing that air pollution is a big problem.

[24] We build an awareness variable that is equal to the google search interest for the pollution-related term ("PM2.5," "PM10," or "air pollution") with the highest value in that country. Countries are assigned to the high awareness subsample if the value of this variable is at or above the median across countries.

$\mathbf{1}\{\Delta T_{c,d} > 0\}$, and (ii) vary the number of city-groups g across which the effects of inversion strength can differ to 15 or 50, from 25 groups used in the baseline specification (Appendix Table A.13). In the second stage, we estimate models where we (i) apply the inverse hyperbolic sine transformation to our measures of output quantity, and (ii) use an alternative, policy-based indicator for unusually high-pollution days as regressor which takes value one when $PM_{2.5}$ exceeds the national air quality standard in the respective country (Table A.14).

We further demonstrate that the results are overall robust to varying the included sets of fixed effects, absorbing common shocks at different geographic and temporal levels, as well as varying the included weather controls (Appendix Figures B.6 to B.7).

A potential concern with the GitHub data is that changes in activity levels in response to pollution might be driven by side projects developers host in public GitHub repos, rather than their core paid work activity. While our descriptive statistics already suggest that the developers in our sample mainly use GitHub for work related coding activities, we further address this concern by repeating the analysis only on the subsample of users who report to work for a company in the current calendar year on their GitHub profile. In this sample of "employees" we find significant effects on output quantity, quality and task choice which are quantitatively very close to the baseline results (Table A.15). This suggests that the drop in code quantity and quality is not driven by side projects of low economic value.

Appendix Table A.16 shows results of regressions using the number of unproductive activities as outcome, which are mostly based on GitHub's social network functions, and do not reflect cognitively-demanding work. We find no significant impact on these activities with either measure of $PM_{2.5}$, and the point estimates imply small absolute and relative changes compared to the effects on productive actions. This implies that the drop in output does not just reflect a shift away from public repos on high-pollution days in favor of other work, e.g. in private repos. In that case, unproductive activities in public repos should be affected as well.

Lastly, we quantify the effect of monthly $PM_{2.5}$ on monthly output. We run 2SLS regressions at the developer×month level, with average monthly $PM_{2.5}$ concentration in $\mu g/m^3$ instrumented with average monthly inversion strength, interacted with indicators for the 25 city-groups. Table A.17 shows that a $1\,\mu g/m^3$ increase in monthly $PM_{2.5}$ reduces the average number of daily actions conducted throughout the month by 0.0025, 0.16% relative to the mean, i.e. the effect magnitude is very close to what we found at the daily level (-0.0021). Again, this effect is driven by reductions in both commits and comments, and the extensive margin contributes roughly 30% to the overall effect. The PR acceptance rate also falls significantly. In sum, air pollution negatively affects developers' output also over more aggregated periods.

7 Effects of Long-Term Pollution Exposure

While the results on acute exposure to $PM_{2.5}$ already show that air pollution generates meaningful economic costs in knowledge intensive sectors, they might provide merely a lower

bound on the potential economic benefit of policies that improve air quality *permanently*. If the effects of exposure to air pollution accumulate over time, prolonged exposure might have a larger effect than what would be implied by just adding up the estimated effect of repeated acute exposure. Identifying causal effects of long-term air pollution exposure, however, is challenging because of the lack of credible instruments that generate long-run variation in air quality. We exploit the broad geographic coverage of our data, and the fact that we can follow developers as they move, even across country boarders, using an inverse movers design similar to Lubczyk and Waldinger (2025). This allows us to provide first evidence on the effect of long-term exposure to air pollution on post-exposure productivity of knowledge workers.

7.1 Inverse Movers Design

The basic idea of the empirical design is to compare the productivity of developers who live in the same city, but have moved there from different origin cities where they were exposed to different levels of $PM_{2.5}$, usually for a prolonged period of time. We then analyse how productivity post-move varies based on pre-move, origin-city $PM_{2.5}$ exposure. The identifying assumption is that origin-city $PM_{2.5}$ is not systematically correlated with other determinants of developer productivity that have a lasting effect even after the developer left the city. To ensure this, we only compare productivity among developers who move to the same destination from different cities in the same country, i.e. we include destination-city × origin-country × year-of-move fixed effects. Furthermore, we control for pre-move productivity and origin city characteristics that might be correlated with air quality, like GDP per capita, climate conditions, and the size of the tech worker population.

Specifically, we estimate

$$Y_i = \beta \, \mathrm{PM}_{o,y} + \theta_{d,c(o),y} + \mathbf{Z}_i'\boldsymbol{\delta} + \boldsymbol{X}_o'\boldsymbol{\gamma} + \varepsilon_i \tag{5}$$

where Y_i denotes average daily actions during the 6 months post-move for developer i who moved to destination city d from origin city o during year y. $PM_{o,y}$ denotes average $PM_{2.5}$ concentration in the origin city during the 12 months preceding the move, as a proxy for long-run pollution exposure. Destination-city × origin-country × year-of-move fixed effects, $\theta_{d,c(o),y}$, ensure that we only exploit variation in air pollution between movers from origin cities in the same country and year, who live in the same destination city post-move. $\mathbf{Z_i}$ contains developers' pre-move productivity, measured by the number of commits during the 12 months preceding the move, and tenure on GitHub at the time of the move, which we both control for flexibly using cubic polynomials. We further add controls for origin-city characteristics, $\boldsymbol{X}_o$, including (i) an indicator for GDP per capita (in 2014$) being above the continent-specific median, (ii) dummies for terciles of the count of GitHub users in our main short-run sample, used as a proxy for local tech-sector size, and (iii) climate conditions, specifically a quadratic function of average daily precipitation and bin variables for average daily temperature during

the 12 months preceding the move.[25] Importantly, since we condition on the destination city, all characteristics of the developer's current location, e.g., post-move contemporaneous air pollution exposure, are controlled for. Standard errors are clustered at the origin city level.

To analyze effect dynamics, i.e. how long the potential effect of prolonged exposure to $PM_{2.5}$ persists, we also estimate a version of the model where we consider daily actions for a period of 12 months post move, broken into quarters t. We then estimate period-specific effects of origin-city $PM_{2.5}$ levels, β^q with $q \in \{1, ..., 4\}$.

$$Y_{it} = \theta_{d,c(o),y} + \theta_{t,d} + \sum_{q=1}^{4} \beta^q \, \mathrm{PM}_{o,y} \cdot \mathbf{1}\{t = q\} + \mathbf{Z}_i'\boldsymbol{\delta} + \mathbf{X}_o'\boldsymbol{\gamma} + \varepsilon_{it} \tag{6}$$

We include the same covariates as above, and further add destination-specific time period fixed effects $\theta_{t,d}$ to capture location-specific time-varying productivity shocks.

7.2 Sample and Descriptives

We observe developers' locations at five points in time in the GHTorrent data: September 25, 2015; April 19, 2016; April 1, 2017; June 1, 2018; and June 1, 2019. As in the short-run analysis, we consider all users who ever made a commit in a professional repository, to focus on professional software developers. We restrict this sample to developers who change their location exactly once during the observation period, either from before to after April 2016; before to after April 2017 or before to after June 2018.[26] We do not know the exact date of the move, but approximate it with the first day we observe the developer at the new location. Moreover, the origin city must be covered by our $PM_{2.5}$ data, i.e., we consider the same 489 cities as in the short-run analysis. There is no restriction on the post-move location, except that it must be a well defined city. We drop moves of less than 200km distance to exclude people who might commute to their new workplace. Lastly, we restrict the sample to developers who had positive activity levels on at least five days during the 12 months preceding the move.

This yields a sample of 3,194 movers. Summary statistics are depicted in Table 7. On average, developers conduct 1.38 daily actions in the year before the move, and 1.30 daily actions during the six months following the move. The average mover has been registered on GitHub since more than 5 years. 49% of developers have reported a company they work for on their profile before the move. This jumps up to 65% post-move, suggesting that a substantial share of location changes is work-related.[27] 4% of developers reported to work for a major tech company pre-move, this also jumps up to 13% after the location change.[28] With a residual SD

[25] We define eight bin variables counting the number of days with average daily temperature in the following ranges (in °C): below -5; [-5,0); [0, 6); [6, 12); [12, 18); [18, 24); [24, 29); [29,). Bin 4 is the reference category.

[26] As the activity data only extends to June 2019, we cannot consider location changes made between June 2018 and June 2019.

[27] These shares do not include developers reporting a university, to exclude students.

[28] Top companies here are defined as the "GAFAM" firms – Google (Alphabet), Amazon, Facebook, Apple, Microsoft – plus Intel, IBM and Uber.

Table 7: Summary statistics

	Mean	SD	SD (residualized)
Developer characteristics			
Pre-move daily actions (1-year mean)	1.38	3.02	2.81
Pre-move actions per active day (1-year mean)	4.54	9.47	7.07
Pre-move daily commits (1-year mean)	0.66	1.16	1.05
Post-move daily actions (6-month mean)	1.30	2.61	2.39
Post-move daily commits (6-month mean)	0.61	1.15	1.05
Github Tenure at move (months)	67.06	20.72	17.95
Employed *post-move*	0.65	0.48	0.43
pre-move	0.49	0.50	0.45
Top company *post-move*	0.13	0.34	0.28
pre-move	0.04	0.20	0.18
Origin-city characteristics			
$PM_{2.5}$ (1-year mean)	14.62	17.04	6.21
$PM_{2.5}$ (2-year mean)	14.90	17.01	5.90
GDP per capita (2014)	56,507	22,526	11,912
Number of professional GitHub Users	507.9	685.3	583.0
Move characteristics			
Distance (km)	3,440	3,580	988
International (%)	31.8	0.47	
Observations			
Number of movers	3194		
Unique origin cities	312		
Unique destination cities	149		

Notes: This table provides summary statistics for the sample of GitHub users who change locations in 2016, 2017 or 2018, and form the analysis sample for estimating the effect of prolonged air pollution exposure. "SD (residualized)" is the standard deviation after residualizing with respect to destination-city × origin-country × year-of-move fixed effects.

of 6.2, there is substantial variation in origin-city $PM_{2.5}$ concentrations, even after controlling for destination-city × origin-country × year-of-move fixed effects. Average distance of moves is 3,440 km, and the majority of moves are domestic.

Appendix Table A.18 depicts the four largest destinations, along with their respective top ten origin cities, illustrating the identifying variation. To estimate the lasting effect of long-term $PM_{2.5}$ exposure, we compare, e.g., performance among developers who have moved to Seattle, investigating whether those coming from cleaner origin cities, such as Austin ($PM_{2.5}$ concentrations of 7.1 $\mu g/m^3$), outperform those coming from more polluted cities in the same country, e.g., Chicago (9.4 $\mu g/m^3$) or Los Angeles (14.4 $\mu g/m^3$).

To test the credibility of the identifying assumption, Appendix Table A.19 shows results from regressions of pre-determined developer and origin-city characteristics on origin-city $PM_{2.5}$, controlling for destination-city × origin-country × year-of-move fixed effects. We test for systematic correlations between origin-city $PM_{2.5}$ and tenure at the time of the move, origin-city GDP per capita, the number of professional GitHub developers in the origin city, the distance between origin and destination, and the probability that developers report an

Table 8: Effects of Long-Run $PM_{2.5}$

	Daily actions			*Total actions*	*Asinh(Daily actions)*
	(1)	(2)	(3)	(4)	(5)
	OLS	OLS	OLS	Poisson	OLS
Origin $PM_{2.5}$ (12m avg)	-0.0091**	-0.0076**		-0.0087*	
	(0.0038)	(0.0033)		(0.0051)	
Log Origin $PM_{2.5}$ (12m avg)			-0.4041**		-0.0992*
			(0.1872)		(0.0529)
Controls					
Pre-move productivity	x	x	x	x	x
Tenure	x		x	x	x
Origin city characteristics	x		x	x	x
Observations	3,194	3,194	3,194	3,192	3,194

Notes: This table reports estimates of β from Equation (5). In the first three columns the dependent variable measures average daily actions during the first six months after a move. Column (4) depicts an estimate from a Poisson regression with the outcome being the count of total actions during the same period. In column (5), the outcome variable is the inverse hyperbolic sine transformation of daily actions. All specifications include destination-city × origin-country × year-of-move fixed effects, and further controls as described at the bottom of the table. Pre-move productivity refers to a cubic polynomial in the number of commits the developer made during the 12 months preceding the move, tenure refers to a cubic polynomial in time since GitHub registration at the time of the move, and origin city characteristics comprise a dummy for origin-city GDP per capita (in 2014$) being above the continent-specific median, three indicators for terciles of the number of professional GitHub users in the origin city and temperature and precipitation controls. Standard errors, clustered at the origin-city level, are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

employer, or a 'top´ employer on their profile in the year before the move. All point estimates are small, and statistically insignificant with one exception. A 1 $\mu g/m^3$ increase in origin $PM_{2.5}$ is associated with an increase in origin GDP per capita in 2014 by 0.6%, indicating that more polluted origin cities are slightly positively selected in terms of local income. Overall, the tests show that developers moving to the same destination from different origin cities in the same country are comparable in terms of observable characteristics.

7.3 Results

Table 8 presents estimates of coefficient β from equation (5). Columns (1) and (2) depict OLS estimates from specifications with the full set of controls and with minimal controls only, respectively. Developers who have been exposed to 1 additional unit of $PM_{2.5}$ in the origin city, on average conduct .0076 to .0091 fewer actions per day during the first six months after the move. Relative to the pre-move sample mean, this implies a 0.55% to 0.66% reduction in actions. These effect sizes are roughly seven times as large as the effect of acute exposure we found in the short-run analysis (-0.083%). In columns (3) to (5) we show that the negative effect of long-term $PM_{2.5}$ exposure does not only arise in the specification with daily actions and $PM_{2.5}$ in levels, but also in a level-log specification, a Poisson model, and when applying the inverse hyperbolic sine transformation to daily actions.

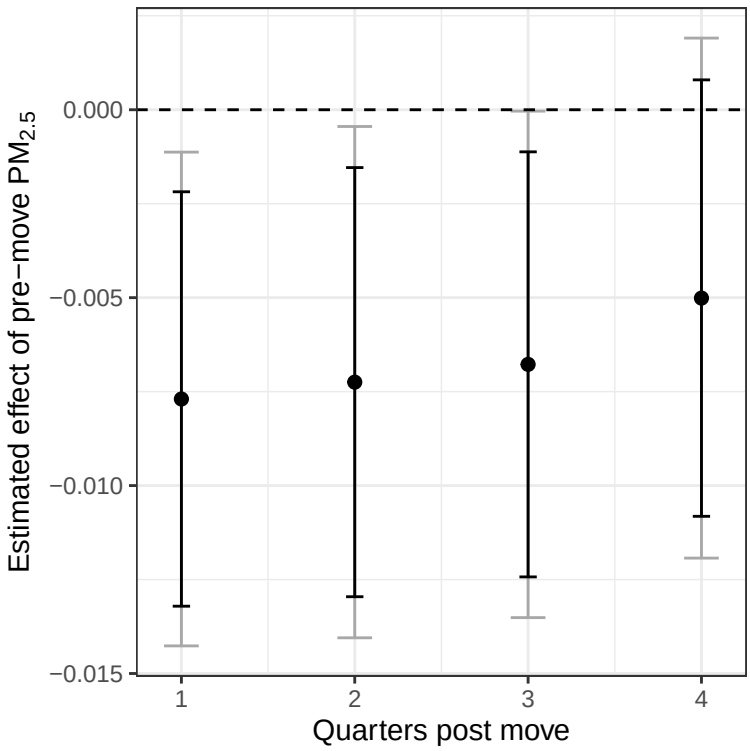


Figure 9: Effect of Long-Term $PM_{2.5}$ Exposure by Quarter Post Move

Notes: Bars plot estimates of coefficients β^q from Equation (6), depicting the effect of pre-move $PM_{2.5}$ on daily actions during the first four quarters q after the move. Vertical lines represent 95%- and 90%-confidence intervals, respectively. The outcome is average daily actions within each quarter. Standard errors are clustered at the origin-city level. All specifications include destination-city × origin-country × year-of-move fixed effects and destination-by-quarter fixed effects, and control variables as in Table 8.

For how long after the move does this adverse effect of prolonged $PM_{2.5}$ exposure persist? While our baseline period is half a year, we next consider a period of 12 months post move[29], and estimate effects separately by quarter. Figure 9 depicts the results. We find negative and significant effects for the first three quarters after the move. The point estimate is largest for the period immediately after the location change, and seems to slowly wear off. By the fourth quarter, the point estimate dropped to 65% the size of the estimate for the first quarter. These estimates likely provide a lower bound on the magnitude and the persistency of the effects of $PM_{2.5}$ exposure since we do not know the exact date of the location change, and developers might already have spent several months in the destination city before the period we consider as the first quarter.

How long is the exposure period underlying these effects? Since we can follow developers back in time only up to September 2015, we cannot directly observe how much time they spent in the origin city, especially for those who moved in 2016. To get a better idea of how long movers lived in the origin city, we consider developers who changed location between June 2018 and June 2019 – a group we can observe for four waves pre-move.[30] Among the 1,844

[29]We cannot extend this period any further because we observe developers' activity only until June 2019, i.e. 12 months after the last date of location change we consider.

[30]This group is not used in the analysis for lack of data on activity post move.

professional users who moved at that time, 65% lived in the origin city at least since April 2016, indicating that origin-city $PM_{2.5}$ captures air pollution exposure over a prolonged period of time.

In Appendix Table A.20 we probe the robustness of the finding that long-term $PM_{2.5}$ exposure reduces post-move performance. The result remains qualitatively and quantitatively similar when (i) considering within-country moves only, (ii) controlling for origin language rather than origin country in the three-way fixed effects to exploit more variation, (iii) controlling for the company developers report on their profiles after the move, (iv) using a two-year average of origin-city $PM_{2.5}$ as regressor and (v) controlling for a broader set of origin-city climate conditions. Finally, we conduct a robustness check where we use commits made between 12 and six months before the assumed date of the move to capture pre-move productivity, instead of commits during the full 12 months. This addresses the issue that we do not observe the exact date of the location change, such that commits a few months before the assumed date could already reflect performance after the move. Moreover, we conduct a placebo test where we randomly assign $PM_{2.5}$ concentration from any of the cities on the developers origin continent to the developer once (Appendix Table A.21) and in 1000 iterations (Appendix Figure B.8).

In Appendix Table A.22 we show that the reduction in total actions is mostly driven by comments, and only to a lesser extent by commits. We also find negative effects on output quality, i.e., the average PR acceptance rate during the six months after the move, which exceeds the effect of acute exposure. However, this result should be taken as suggestive evidence only as it is based on a smaller number of developers (862) who opened a minimum number of PRs pre- and post-move.

These results complement the findings by Lubczyk and Waldinger (2025), who find no significant long-run effects of pollution exposure on wages for individuals exposed as adults. Using our measures of developer output quantity and quality, we can analyze more subtle productivity effects that might not show up in wages.

Effect Magnitudes. The results allow us to compare the causal impact of acute and long-term exposure to $PM_{2.5}$ in the same context. The elasticity of daily output w.r.t. long-term $PM_{2.5}$ exposure is -0.1 (Table 8, Column 5), which is an order of magnitude larger than the elasticity of daily output w.r.t. acute $PM_{2.5}$ exposure, which we estimate to be -0.0125 based on the short-run estimate in Table 2. This indicates that long-term exposure to $PM_{2.5}$ generates substantially larger adverse impacts on productivity, which persist for at least three quarters after the change of location – pointing towards even larger economic damages from air pollution than the short-run estimates would suggest. Comparing the short- and long-run elasticities to results on the effect of acute air pollution exposure on productivity or performance in other contexts, we find that our short-run elasticity estimate is at the lower end of effect sizes found (Figure 10). The effect on developers' output is in particular much smaller than estimates obtained for

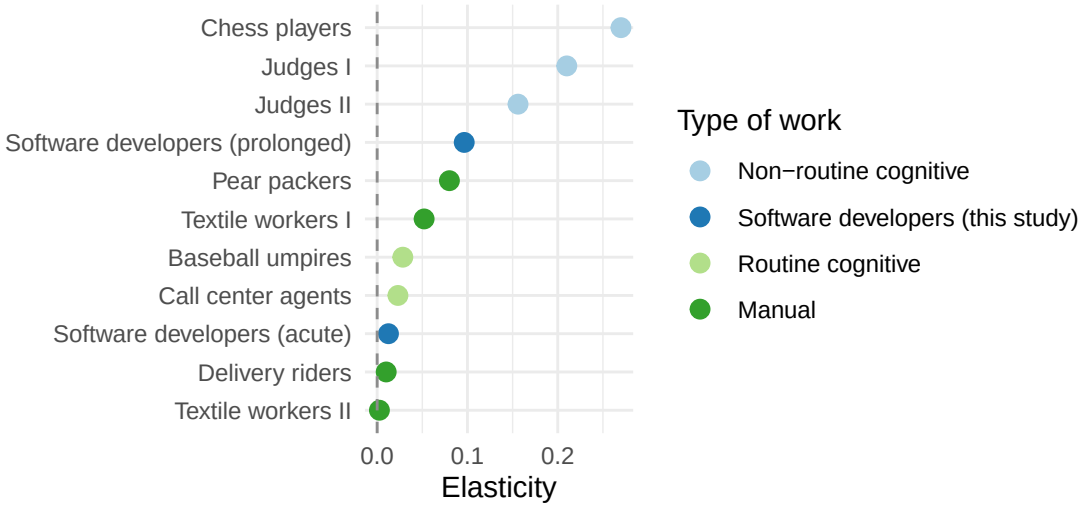


Figure 10: Effects of air pollution across occupations

Notes: Elasticities for our study are based on Tables 2 and 8. Elasticities from other studies: Künn et al. (2023) (chess players), Kahn and Li (2020) (judges I), Sarmiento (2022) (judges II), Chang et al. (2019) (call center agents), Adhvaryu et al. (2022) (textile workers I), He et al. (2019) (textile workers II), Chang et al. (2016) (pear packers), D'Adda et al. (2025) (delivery riders), and Archsmith et al. (2018) (baseball umpires).

judges and chess players, who also engage in cognitively demanding non-routine tasks. A potential explanation is that chess tournaments and court hearings are highly inflexible settings that do not allow for adapting working hours or the choice of tasks to productivity shocks. This contrasts with our setting where workers can to some extent adjust to a short-run increase in $PM_{2.5}$. While the elasticity for prolonged exposure is substantially larger than the short-run elasticity, it is still below the estimates of short-run elasticities found in less flexible knowledge intensive settings. This underscores the importance of our analysis: It might be misleading to estimate the total economic cost of air pollution, or other environmental shocks, based on inflexible settings given that most knowledge workers have at least some degree of flexibility in organizing their workday.

8 Conclusion

How do environmental conditions affect workers in jobs that form the backbone of the modern knowledge economy? In our paper, we provide insights on this question, which is highly relevant as such jobs are expected to become even more widespread as digitalization and automation continue to change the world of work (OECD, 2026). Using rich data on GitHub activities, we show that air pollution reduces daily output quantity and quality in a global sample of software developers. Developers exploit the flexibility of their work setting to adapt to $PM_{2.5}$ shocks by shifting to less complex tasks and reallocating work from high-pollution, low-productivity workdays to low-pollution, high-productivity weekends. These adjustments likely help to alleviate effects on output quantity and quality, but they also imply additional welfare costs not captured by changes in output due to forgone leisure time on weekends and

negative impacts on work-life balance. While our estimates are at the lower end of air pollution effects found in less flexible and less collaborative occupations, the implied annual loss of $266 per worker due to high pollution days is economically relevant, due to the high value generated by knowledge workers. Using an inverse movers design, we find that prolonged exposure to $PM_{2.5}$ has a substantially larger negative effect on output than short-run exposure, which persists for at least three quarters post exposure, pointing to even higher aggregate productivity costs of air pollution.

While we use data on software developers using GitHub as part of their formal work, we believe that the findings generalize to other occupations that offer flexible schedules and discretion in task choice and require similar skills like problem-solving, attention to detail, programming, and teamwork. This applies to many knowledge workers, including business analysts or researchers.

Our findings have important policy implications. When setting air pollution standards, regulators should factor in the growing evidence on the economic benefits of pollution reductions through productivity gains. Ignoring such benefits—for example, productivity gains among knowledge workers—will bias cost–benefit analyses downward and may yield standards that are too lenient. Importantly, we find adverse effects of $PM_{2.5}$ on output even below the current regulatory standards in the European Union and the US. An interesting avenue for future research might be to investigate whether the adoption of generative AI tools by knowledge workers alters the air pollution-productivity relationship.

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Appendix (For Online Publication)

A Additional Tables

Table A.1: Labels Indicating *Easy* Issues

good first issue	good first bug	good-first
documentation	polish	cleanup
simple	easy	small
trivial	minor	**help wanted**
junior job	newcomer	starter
beginner	newbie	novice
low hanging	low-hanging	

Notes: If a label contains any of these terms, the issue is classified as "easy". Bolt text indicates GitHub default labels.

Table A.2: Description of Outcome Variables

Domain	Concept	Variable	Details
Output Quantity	Total output quantity	Actions	Sum of number of commits, comments on issues, PRs and commits, PRs opened, PRs closed, issues opened, closed and reopened
	Coding activity	Commits	Number of commits
	Interactive activity	Comments	Sum of number of comments written on issues, PRs and commits
	Extensive Margin	Any actions	at least one activity out of those mentioned above was conducted
Output Quality	Proposed code changes that get accepted	PR Acceptance Rate	PRs opened that got merged / PRs opened that got closed
Task choice	Average PR complexity	Lines added per PR	Average number of lines of code added in PRs opened or closed
		Files changed per PR	Average number of code files changed in PRs opened or closed
		Lines deleted per PR	Average number of lines of code deleted in PRs opened or closed
	Easy tasks among issue events	Share easy issue events	(#easy issues opened + #easy issues closed + #comments written on easy issues)/ (#issues opened + #issues closed + #comments written on issues)
Working hours	Same day working hours	Time last action	Minute of final action of the day
		Time first action	Minute of first action of the day
	Weekend Activity	Time first/last action	Average minute of first/final action on weekends

Notes: The Table displays information on the outcome variables we use, how they are constructed, and what they measure.

Table A.3: Summary Statistics for working days and non-working days

	Mean	SD	SD (within)	Min	Max	Observations
Working Days						
Actions	3.023	7.48	6.41	0	291	13,407,752
Any action	0.4	0.49	0.43	0	1	13,407,752
Actions | Actions > 0	7.557	10.27	8.94	1	291	5,364,119
Time first action	12:47	4.78 hours	4.34 hours	3:00	3:00	5,346,867
Time last action	17:18	4.95 hours	4.55 hours	3:00	3:00	5,346,867
Non-Working Days						
Actions	1.429	5.09	4.65	0	290	6,168,559
Any action	0.226	0.42	0.38	0	1	6,168,559
Actions | Actions > 0	6.322	9.16	8.36	1	290	1,394,123
Time first action	14:19	5.43 hours	5.02 hours	3:00	3:00	1,385,657
Time last action	18:11	5.35 hours	4.95 hours	3:00	3:00	1,385,657

Notes: The Table displays summary statistics separately for weekdays (upper panel) and weekends and public holidays (bottom panel).

Table A.4: Sources of air quality data

Geographic area	Data source
United States	U.S. Environmental Protection Agency
Canada	Canadian National Air Pollution Surveillance Program
Mexico	Gobierno de Jalisco; Gobierno de la Ciudad de México
Brazil	Instituto de Energia e Meio Ambiente
Montevideo (Uruguay)	Intendencia de Montevideo
Austria, Belgium, Bulgaria, Croatia, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland, United Kingdom	European Environment Agency AirBase
Belfast (United Kingdom), Cambridge (United Kingdom)	UK Department for Environment, Food and Rural Affairs
Armenia, Belarus, Bosnia and Herzegovina, Egypt, Georgia, Israel, Jordan, Malta, Moldova, North Macedonia, Russia, Serbia, Türkiye, Ukraine	Copernicus Atmosphere Monitoring Service
China	China National Environmental Monitoring Centre
Bengaluru (India)	Central Pollution Control Board
Chennai (India), Dhaka (Bangladesh), Hanoi (Vietnam), Ho Chi Minh City (Vietnam), Hyderabad (India), Jakarta (Indonesia), Kolkata (India), Mumbai (India), New Delhi (India)	U.S. Embassies, AirNow.gov
Hong Kong	Hong Kong Environmental Protection Department
Japan	National Institute for Environmental Studies
Singapore	National Environment Agency
South Korea	Air Korea
Taiwan	Environmental Protection Administration
Australia	ACT Government; New South Wales Department of Planning and Environment; Queensland Government; South Australian Government; Victorian Government
New Zealand	Stats NZ Tatauranga Aotearoa

Table A.5: Developer-by-date observations and inversion frequency by geographic regions R

Region R	Observations	Share of Total (%)	Inversion Frequency (%)
Eastern Asia	2,599,202	13.3	32
Eastern Europe	1,210,309	6.2	57
Northern America	8,796,074	44.9	51
Northern Europe	2,121,298	10.8	35
Oceania	564,666	2.9	37
South America	191,842	1.0	42
South-Eastern Asia	197,078	1.0	4
Southern Asia	386,747	2.0	36
Southern Europe	563,528	2.9	49
Western Asia	153,764	0.8	45
Western Europe	2,791,803	14.3	51

Notes: The table shows the distribution of observations in the developer × date panel described in section 3.2 across geographic regions R (Columns (2) and (3)). In Column (4), it shows the share of all city × date observations with a nighttime inversion by geographic region R.

Table A.6: Effect of $PM_{2.5}$ on Quantity of Issue and Pull Request Actions

	PRs closed (1)	*PRs opened* (2)	*Issues closed* (3)	*Issues opened* (4)
Panel A: Levels				
$PM_{2.5}$	-0.0001**	-0.00010**	-0.00007*	-0.0002***
	(0.00006)	(0.00004)	(0.00004)	(0.00003)
First stage F-stat.	171.3	171.3	171.3	171.3
% change	-0.087	-0.073	-0.069	-0.191
Panel B: Shocks				
$PM_{2.5}$ shock	-0.0046	-0.0039	-0.0003	-0.0080***
	(0.0059)	(0.0044)	(0.0037)	(0.0027)
First stage F-stat.	110.3	110.3	110.3	110.3
% change	-3.10	-2.89	-0.259	-8.08
Observations	856,522	856,522	856,522	856,522
Dependent variable mean	0.147	0.135	0.107	0.099

Notes: 2SLS estimates of the parameter β in Equation (1). The regressor of interest is $PM_{2.5}$ concentration in $\mu g/m^3$ (Panel A) or a binary $PM_{2.5}$ shock variable as in Equation (3) (Panel B). Covariates include fixed effects and flexible weather controls as described in Table 2. Regressions are weighted by the number of active workers in a city-month cell. Standard errors clustered at the city level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.7: OLS Results for Work Quantity and Quality

	Actions (1)	*Any actions* (2)	*PR accept. rate* (3)	*Commits* (4)	*Comments* (5)	*Actions* (6)	*Any actions* (7)	*PR accept. rate* (8)	*Commits* (9)	*Comments* (10)
Panel A: Levels										
$PM_{2.5}$	-0.0005***	-0.00003**	-0.00002	-0.0002***	-0.0002**	-0.0006***	-0.00003**	-0.00006	-0.0003***	-0.0001**
	(0.0001)	(0.00001)	(0.00003)	(0.00006)	(0.00008)	(0.0002)	(0.00001)	(0.00004)	(0.00009)	(0.00007)
Panel B: Shocks										
$PM_{2.5}$ shock	-0.0102	-0.0007	0.0010	-0.0036	-0.0048	-0.0131	-0.0008	-0.0005	-0.0053	-0.0049
	(0.0089)	(0.0007)	(0.0014)	(0.0044)	(0.0033)	(0.0094)	(0.0006)	(0.0012)	(0.0046)	(0.0038)
Fixed effects										
City	✓	✓	✓	✓	✓					
Country × month	✓	✓	✓	✓	✓					
Region × day-of-week	✓	✓	✓	✓	✓					
Region × year-month	✓	✓	✓	✓	✓					
City × month						✓	✓	✓	✓	✓
Region × date						✓	✓	✓	✓	✓
Observations	856,522	856,522	334,060	856,522	856,522	855,904	855,904	332,616	855,904	855,904

Notes: OLS estimates of the parameter β in Equation (1). The regressor of interest is $PM_{2.5}$ concentration in $\mu g/m^3$ (Panel A) or a binary $PM_{2.5}$ shock variable as in Equation (3) (Panel B). Covariates: flexible weather controls as described in Section 4. Included fixed effects are displayed in the bottom part of the table. Regressions are weighted by the number of active workers in a city-month cell. Standard errors clustered at the city level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.8: Lagged Effects of Inversion Strength

	Actions (1)	*Any actions* (2)	*PR acceptance rate* (3)	*Commits* (4)	*Comments* (5)
Inversion strength (t)	-0.0033***	-0.0002**	-0.0004**	-0.0016***	-0.0010
	(0.0012)	(0.00010)	(0.0002)	(0.0005)	(0.0006)
Inversion strength (t-1)	0.0004	0.00005	-0.00009	0.00006	-0.00004
	(0.0011)	(0.00008)	(0.0001)	(0.0005)	(0.0004)
Inversion strength (t-2)	-0.0003	-0.00001	-0.0002*	-0.0001	-0.0002
	(0.0007)	(0.00006)	(0.0001)	(0.0003)	(0.0003)
Inversion strength (t-3)	0.0008	0.00002	-0.0003**	0.0004	0.0001
	(0.0010)	(0.00006)	(0.0001)	(0.0005)	(0.0004)
Observations	844,592	844,592	330,976	844,592	844,592

Notes: OLS estimates of the outcomes displayed at the top on inversion strength in degrees Celsius on the same day, as well as three lags. Covariates: as described in Table 2. Regressions are weighted by the number of active workers in a city-month cell. Standard errors clustered at the city level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.9: Effect of $PM_{2.5}$ on PRs opened and closed with GHArchive and GHTorrent data

	PRs opened (GHA) (1)	*PRs closed (GHA)* (2)	*PRs opened (GHT)* (3)	*PRs closed (GHT)* (4)
$PM_{2.5}$	-0.0001***	-0.0001***	-0.00009**	-0.0001*
	(0.00004)	(0.00005)	(0.00004)	(0.00006)
Observations	725,185	725,185	725,185	725,185

Notes: 2SLS estimates of the parameter β in equation (1), where the outcome is the number of pull requests (PRs) opened or closed, respectively. This is measured in GHArchive data in columns (1) to (2) and GHTorrent data in column (3) to (4). The regressor of interest is $PM_{2.5}$ concentration in $\mu g/m^3$. Covariates are as described in Table 2. The sample period is 2015 to May 2019. Regressions are weighted by the number of active workers in a city- month cell. Standard errors clustered at the city level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.10: Effect of workweek $PM_{2.5}$ on Work Activity on High-Pollution Weekends

	Time of first action		*Time of last action*	
	(1)	(2)	(3)	(4)
Days with $PM_{2.5}$ shock	0.7928		-1.260	
	(2.228)		(2.472)	
Any $PM_{2.5}$ shock		3.559		0.6822
		(6.195)		(6.427)
First stage F-stat.	24.1	26.2	24.1	26.2
Observations	269,704	269,704	269,704	269,704

Notes: 2SLS estimates of the parameter β in equation 4. 2SLS regressions are estimated at the developer × week level. Models are estimated only on weeks with PM2.5 above the city-specific 75th percentile on at least one weekend day. Standard errors clustered at the city level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.11: Placebo Test: Effect of $PM_{2.5}$ Wednesday to Sunday on Work Activity Monday to Tuesday

	Actions		*PR complexity*		*Time of first action*		*Time of last action*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Days with $PM_{2.5}$ shock	-0.0797		-0.0056		1.416		-1.172	
	(0.0818)		(0.0087)		(1.411)		(1.758)	
Any $PM_{2.5}$ shock		-0.1033		-0.0119		3.811		-1.485
		(0.1984)		(0.0202)		(3.066)		(4.084)
First stage F-stat.	40.4	38.2	30.4	32.0	45.6	27.2	45.6	27.2
Observations	1,847,551	1,847,551	332,870	332,870	960,287	960,287	960,287	960,287

Notes: 2SLS estimates of the parameter β in a placebo version of equation 4. 2SLS regressions are estimated at the developer × week level. Outcomes refer to Monday and Tuesday, the placebo weekend. The regressor of interest is the number of days characterized by a $PM_{2.5}$ shock during the placebo workweek, i.e. the previous five days, or an indicator for at least one $PM_{2.5}$ shock during the placebo workweek. Models are estimated only on weeks with PM2.5 below the city-specific 75th percentile on both placebo weekend days. Covariates: developer and region-by-year-by-quarter fixed effects, third-order polynomial of developer experience, number of public holidays during the workweek, leads of the instrumental variables for the placebo weekend, flexible weather controls for the placebo exposure period and the placebo weekend. Standard errors clustered at the city level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.12: Heterogeneity: Awareness and Building Stock

		Actions	
	Full sample	Low	High
	(1)	(2)	(3)
Panel A: Google Trends.			
$PM_{2.5}$	-0.0021***	-0.0047***	-0.0018***
	(0.0006)	(0.0014)	(0.0005)
Observations	846,733	385,180	461,553
First stage F-stat.	171.2	930.9	191.3
Mean $PM_{2.5}$	15.10	12.07	17.06
Mean dep. var.	2.5136	2.4842	2.5327
Google search intensity index (0-1)	0.35	0.17	0.46
Panel B: Pew Research.			
$PM_{2.5}$	-0.0025***	-0.0060***	-0.0020***
	(0.0008)	(0.0020)	(0.0007)
Observations	684,101	158,874	525,227
First stage F-stat.	353.9	151.3	367.1
Mean $PM_{2.5}$	11.89	10.99	12.16
Mean dep. var.	2.6553	2.6149	2.6672
Share *"air pollution is big problem"*	65.3%	52.9%	69.0%
Panel C: Building Stock Age.			
$PM_{2.5}$	-0.0045*	-0.0005	-0.0097***
	(0.0023)	(0.0025)	(0.0023)
Observations	574,033	284,191	289,842
First stage F-stat.	153.4	143.1	224.6
Mean $PM_{2.5}$	10.33	10.82	9.99
Mean dep. var.	2.7647	2.8297	2.7192
Share old buildings	33.1%	8.6%	50.4%
Share modern buildings	12.8%	13%	12.6%

Notes: Table depictes 2SLS estimates. In Panel A, B, and C, respectively, the estimates in Column (1) are based on all cities covered by the google trends data, the Pew Research Center International Science Survey, and the building stock data, respectively. Results in Columns (2) and (3) are estimated from subsamples with below and above median values of air pollution awareness (Panels A and B) or share of old dwellings (Panel C). The precise variables used for the sample splits are google search interest for air pollution related terms (Panel A), the share of respondents stating that air pollution is a big problem in the Pew Survey (Panel B), and the share of dwellings built before 1960 (Panel C), which are defined as old buildings. Modern buildings are those built after 2000. Regressions weighted by number of developers in city-month cell. Standard errors, clustered by city, in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.13: Robustness: First Stage Specification

	Actions (1)	*Any action* (2)	*PR acceptance rate* (3)	*Commits* (4)	*Comments* (5)	*PR complexity* (6)	*Share easy issue events* (7)
Panel A: $PM_{2.5}$—IV: Inversion Indicator × 25 city groups.							
$PM_{2.5}$	-0.0036***	-0.0002***	-0.0002**	-0.0016***	-0.0012***	-0.0008**	-2.08×10^{-6}
	(0.0006)	(0.00005)	(0.0001)	(0.0003)	(0.0003)	(0.0003)	(0.00004)
First stage F-stat.	100.2	100.2	101.2	100.2	100.2	98.5	94.9
Panel B: $PM_{2.5}$ shock—IV: Inversion Indicator × 25 city groups.							
$PM_{2.5}$ shock	-0.1353**	-0.0078	0.0021	-0.0830***	-0.0428	-0.0148	-0.0022
	(0.0613)	(0.0050)	(0.0080)	(0.0279)	(0.0277)	(0.0133)	(0.0018)
First stage F-stat.	60.8	60.8	59.4	60.8	60.8	63.6	59.2
Observations	856,522	856,522	334,060	856,522	856,522	374,588	562,922
Panel C: $PM_{2.5}$—IV: Inversion strength × 15 city groups.							
$PM_{2.5}$	-0.0022***	-0.0002***	-0.0005***	-0.0010***	-0.0007**	-0.0006***	0.00002
	(0.0006)	(0.00005)	(0.0001)	(0.0003)	(0.0003)	(0.0002)	(0.00003)
First stage F-stat.	199.4	199.4	253.0	199.4	199.4	252.0	215.1
Panel D: $PM_{2.5}$ shock—IV: Inversion strength × 15 city groups.							
$PM_{2.5}$ shock	-0.1425***	-0.0095**	-0.0162**	-0.0725***	-0.0442*	-0.0279***	-0.0002
	(0.0522)	(0.0038)	(0.0067)	(0.0219)	(0.0246)	(0.0104)	(0.0011)
First stage F-stat.	95.6	95.6	80.7	95.6	95.6	97.7	96.9
Observations	856,522	856,522	334,060	856,522	856,522	374,588	562,922
Panel E: $PM_{2.5}$—IV: Inversion strength × 50 city groups.							
$PM_{2.5}$	-0.0021***	-0.0002***	-0.0004***	-0.0010***	-0.0007**	-0.0006***	0.00002
	(0.0006)	(0.00005)	(0.0001)	(0.0003)	(0.0003)	(0.0002)	(0.00003)
First stage F-stat.	172.9	172.9	188.4	172.9	172.9	177.1	176.4
Panel F: $PM_{2.5}$ shock—IV: Inversion strength × 50 city groups.							
$PM_{2.5}$ shock	-0.1082**	-0.0072*	-0.0136**	-0.0613***	-0.0304	-0.0201**	-0.0003
	(0.0548)	(0.0039)	(0.0067)	(0.0225)	(0.0257)	(0.0093)	(0.0011)
First stage F-stat.	180.4	180.4	127.5	180.4	180.4	142.1	182.9
Observations	856,522	856,522	334,060	856,522	856,522	374,588	562,922

Notes: 2SLS estimates of the parameter β in Equation (1). In Panels A, C and E, the regressor of interest is $PM_{2.5}$ concentration in $\mu g/m^3$. In Panel B, D and F, an indicator for a $PM_{2.5}$ shock is used instead. Relative to specifications underlying results in Table 2, the first stage model is changed: In Panels A and B, instruments are the interactions of a dummy variable for a thermal inversion occurring with indicators for 25 distinct city groups g. In Panels C to F, the first stage specification is as in Equation (2), but we form 15 city-groups g (C and D) or 50 city-groups g (E and F) instead of 25 groups (baseline). Covariates as described in Table 2. Regressions are weighted by the number of active workers in a city during the current month. Standard errors clustered at the city level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.14: Robustness: Second Stage Specification

	Actions Asinh (1)	*Commits Asinh* (2)	*Comments Asinh* (3)
Panel A: $PM_{2.5}$—Inv. hyperbolic sine transformation.			
$PM_{2.5}$	-0.0005***	-0.0003***	-0.0002***
	(0.0001)	(0.00007)	(0.00008)
First stage F-stat.	171.3	171.3	171.3
Observations	856,522	856,522	856,522
Panel B: $PM_{2.5}$ shock—Inv. hyperbolic sine transformation.			
$PM_{2.5}$ shock	-0.0215**	-0.0148**	-0.0100
	(0.0101)	(0.0064)	(0.0066)
First stage F-stat.	110.3	110.3	110.3
Observations	856,522	856,522	856,522

Panel C: $PM_{2.5}$ exceeding national standard.

	Actions	*Any actions*	*PR acceptance rate*	*Commits*	*Comments*	*PR complexity*	*Share easy issue events*
$PM_{2.5}$ > standard	-0.1033**	-0.0079**	-0.0123**	-0.0502**	-0.0426*	-0.0163**	-0.0011
	(0.0505)	(0.0040)	(0.0061)	(0.0210)	(0.0244)	(0.0078)	(0.0009)
First stage F-stat.	90.6	90.6	77.7	90.6	90.6	99.2	88.6
Observations	758,955	758,955	291,894	758,955	758,955	328,976	496,226

Notes: 2SLS estimates of the parameter β in Equation (1). In Panel A, the regressor of interest is $PM_{2.5}$ concentration measured in $\mu g/m^3$. In Panel B, an indicator for a $PM_{2.5}$ shock is used instead (as defined in Equation 3). In Panel C, the regressor is a binary indicator for $PM_{2.5}$ concentration violating the national air quality standard. The first stage specification is given in Equation (2). Covariates as described in Table 2. Regressions are weighted by the number of active workers in a city during the current month. Standard errors clustered at the city level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.15: Results for software developers reporting a company

	Actions (1)	*PR acceptance rate* (2)	*PR complexity* (3)	*Share easy issue events* (4)
$PM_{2.5}$	-0.0025***	-0.0004***	-0.0004*	0.00006*
	(0.0007)	(0.0001)	(0.0002)	(0.00003)
Dependent variable mean	2.73	0.670	0.019	0.029
% change	-0.091	-0.054	-2.26	0.209
First stage F-stat.	154.5	229.5	215.3	148.8
Observations	13,224,348	1,012,029	1,468,329	2,899,938

Notes: Regressions are estimated at the developer × date level using two-stage least squares. Developers are only included in the sample, if they report an employer on their GitHub profile in the respective calendar year. Users reporting no employer or a university are dropped. Covariates: developer, region-by-day-of-week, country-by-month, and region-by-year-by-month fixed effects, cubic polynomial in developers' time since registration on GitHub as well as weather covariates as described in Section 4. Standard errors, clustered by city, in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.16: Placebo Test: Effect of $PM_{2.5}$ on Unproductive Actions (IV Estimates)

	Unproductive Actions	
	(1)	(2)
$PM_{2.5}$	-0.0002	
	(0.0001)	
$PM_{2.5}$ shock		-0.0090
		(0.0111)
First stage F-stat.	171.3	110.3
Observations	856,522	856,522
Dependent variable mean	0.527	0.527
% change	-0.030	-1.70

Notes: 2SLS estimates of the parameter β in Equation (1). The dependent variable measures the number of unproductive actions conducted per day (following other users, watching repositories, creating new repositories, (un)subscribing to issues, (un)assigning users to issues, labeling issues). Covariates include fixed effects and flexible weather controls as described in Table 2. Regressions are weighted by the number of active workers in a city-month cell. Standard errors clustered at the city level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.17: Analysis at the Monthly Level (IV Estimates)

	Daily actions	*Days with actions*	*PR acceptance rate*	*Daily commits*	*Daily comments*
	(1)	(2)	(3)	(4)	(5)
$PM_{2.5}$ (month)	-0.0025***	-0.0092***	-0.0004***	-0.0008*	-0.0010**
	(0.0009)	(0.0026)	(0.0001)	(0.0004)	(0.0004)
First stage F-stat.	20.2	20.2	24.0	20.2	20.2
Observations	641,495	641,495	332,039	641,495	641,495
Dependent variable mean	2.53	10.5	0.634	1.18	0.850

Notes: The regressor is average monthly $PM_{2.5}$ in $\mu g/m^3$, instrumented by average monthly night-time inversion strength, interacted with city group indicators. Dependent variables in Columns (1), (4) and (5) measure average daily activities at the developer-by-month level. Covariates: Developer, country-by-month, and region-by-year-by-month fixed effects, cubic polynomial of developer's experience on GitHub, eight variables counting the number of days during the month falling into different temperature bins, third-order polynomials in average monthly precipitation, relative humidity, and wind speed, number of holidays and days with heavy wildfire smoke. Effects of weather controls and holidays can vary across regions. Standard errors, clustered by city, in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.18: Top origin-destination pairs

San Francisco, US			New York, US			Seattle, US			London, GB		
Origin	N	$PM_{2.5}$	Origin	N	$PM_{2.5}$	Origin	N	$PM_{2.5}$	Origin	N	$PM_{2.5}$
New York, US	56	7.58	S. Francisco, US	58	8.41	S. Francisco, US	31	9.30	Berlin, DE	17	14.98
Boston, US	26	6.80	London, GB	16	10.97	Chicago, US	17	9.43	Paris, FR	11	13.62
Cambridge, US	24	6.83	Washington, US	16	8.69	Austin, US	12	7.08	Edinburgh, GB	8	5.63
LA, US	24	14.48	Boston, US	13	6.92	New York, US	9	7.47	Manchester, GB	8	9.85
Toronto, CA	24	7.72	Seattle, US	13	6.77	LA, US	7	14.42	Singapore, SG	8	17.95
London, GB	20	10.47	Cambridge, US	12	7.16	Toronto, CA	7	7.76	Sydney, AU	7	8.04
Atlanta, US	18	9.98	Austin, US	10	7.53	Ann Arbor, US	6	8.03	Munich, DE	6	11.99
Paris, FR	18	14.32	Paris, FR	10	14.63	Boston, US	6	7.16	Moscow, RU	5	17.63
Pittsburgh, US	17	9.88	Chicago, US	9	9.38	San Diego, US	6	11.55	New York, US	5	7.60
Washington, US	17	9.06	Research Triangle Park, US	8	9.38	Washington, US	6	9.02	Tokyo, JP	5	14.15

Notes: The table reports the four most frequent destination cities in the sample of professional GitHub users who change location in 2016, 2017 or 2018; along with their ten most frequent origin cities. *N* depicts the number of sample developers who move between the respective origin-destination pair and $PM_{2.5}$ reports the average concentration of fine particulate matter in the origin city. These numbers pool across developers moving in different years.

Table A.19: Falsification Tests: Origin $PM_{2.5}$ and pre-determined Origin and Developer Characteristics

	Tenure (log) (1)	*Pro GitHub users (log)* (2)	*GDP per cap. (log)* (3)	*Distance (log)* (4)	*At company* (5)	*At top company* (6)
Panel A. Levels						
Origin $PM_{2.5}$ (12m avg)	0.0009	0.0075	0.0064***	0.0053	-0.0018	-0.0005
	(0.0007)	(0.0083)	(0.0021)	(0.0038)	(0.0011)	(0.0004)
Panel B. Logs						
Log Origin $PM_{2.5}$ (12m avg)	0.0231	0.2603	0.0750	-0.1134	-0.0260	-0.0378
	(0.0361)	(0.5007)	(0.1054)	(0.2236)	(0.0558)	(0.0334)
Observations	3,194	3,194	3,194	3,194	3,194	3,194

Notes: This table reports estimates from regressions of pre-determined developer and origin city characteristics on average $PM_{2.5}$ concentration in the origin city during the 12 months preceding the developer's location change, or the log thereof. Each coefficient comes from a different regression. All regressions control for destination city × year of move × origin country fixed effects. The dependent variables are given by the log of the developer's tenure on GitHub at the time of the move, the log of the number of GitHub users in our main short-run sample who reside in the origin city, the origin city's 2014 GDP per capita ($), the log of the distance between origin and destination city, a dummy for the developer reporting a company on their profile before the move, and a dummy for the developer reporting a 'top' tech company before the move (Google, Apple, Facebook, Amazon, Microsoft, Intel, IBM, Uber). Standard errors, clustered at the origin-city level, are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.20: Long Term Exposure: Robustness Tests

	Daily actions						
	Internal moves only (1)	Origin language (2)	Post company group (3)	Post company (4)	24 months $PM_{2.5}$ (5)	Extended climate controls (6)	Productivity pre-move (7)
$PM_{2.5}$ (12m avg)	-0.0074*	-0.0086**	-0.0089**	-0.0136*		-0.0075**	-0.0088*
	(0.0044)	(0.0035)	(0.0039)	(0.0081)		(0.0037)	(0.0045)
$PM_{2.5}$ (24m avg)					-0.0071**		
					(0.0034)		
Observations	2,178	3,186	3,194	1,846	3,194	3,194	3,194

Notes: This table reports estimates of β from Equation (5). The dependent variable measures average daily actions during the first six months after a move. Relative to the baseline specification in Column (1) of Table 8, the following aspects are varied: In Column (1) the sample is reduced by dropping international movers. In Column (2), the destination-by-origin country-by-year fixed effect is replace by a destination-by-origin language-by-year fixed effect and an origin country fixed effect. In Column (3), three dummy variables are added to control for the type of company the developer reports on their profile post move (no company, university, GAFAM+, or another large company among the top 20 employers in the sample). In Column (4), a fixed effect for company-by-world region post move is added. In Column (5), origin city PM2.5 concentration averaged over 24 months - rather than 12 months - preceding the move is used as regressor, and climate controls are measured over the same period. In Column (6), quadratic functions of origin city wind speed and relative humidity are added. In Column (7) pre move productivity is measured by the number of commits made 12 to six months before the assumed date of the location change, rather than the full year. Standard errors, clustered at the origin-city level, are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.21: Long-Term Exposure: Placebo Test

	Daily actions	
	(1)	(2)
Origin $PM_{2.5}$ (12m avg)	-0.0091**	
	(0.0038)	
Origin $PM_{2.5}$ (placebo)		-0.0021
		(0.0027)
Observations	3,194	3,194

Notes: Column (1) reproduces Column (1) of Table 8, the main estimate of β from Equation (5). Column (2) reports results from a placebo test, where PM2.5 is randomly reassigned across origin cities within move-years × continent cells. Standard errors, clustered at the origin-city level, are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A.22: Long-Term Exposure: Further Outcomes

	PR acceptance rate	*Daily commits*	*Daily comments*
	(1)	(2)	(3)
Origin $PM_{2.5}$	-0.0021**	-0.0026	-0.0043**
	(0.0010)	(0.0016)	(0.0018)
Observations	862	3,194	3,194

Notes: This table reports estimates of β from Equation (5), for three distinct outcome variables given at the top of the table. In Column (1) the sample size is reduced as we consider only developers who opened more than three PRs post move and more than six PRs pre move, i.e. on average at least one PR every other month. In Column (1) instead of pre-move commits, we use the pre-move PR acceptance rate to control for pre-move productivity. Standard errors, clustered at the origin-city level, are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

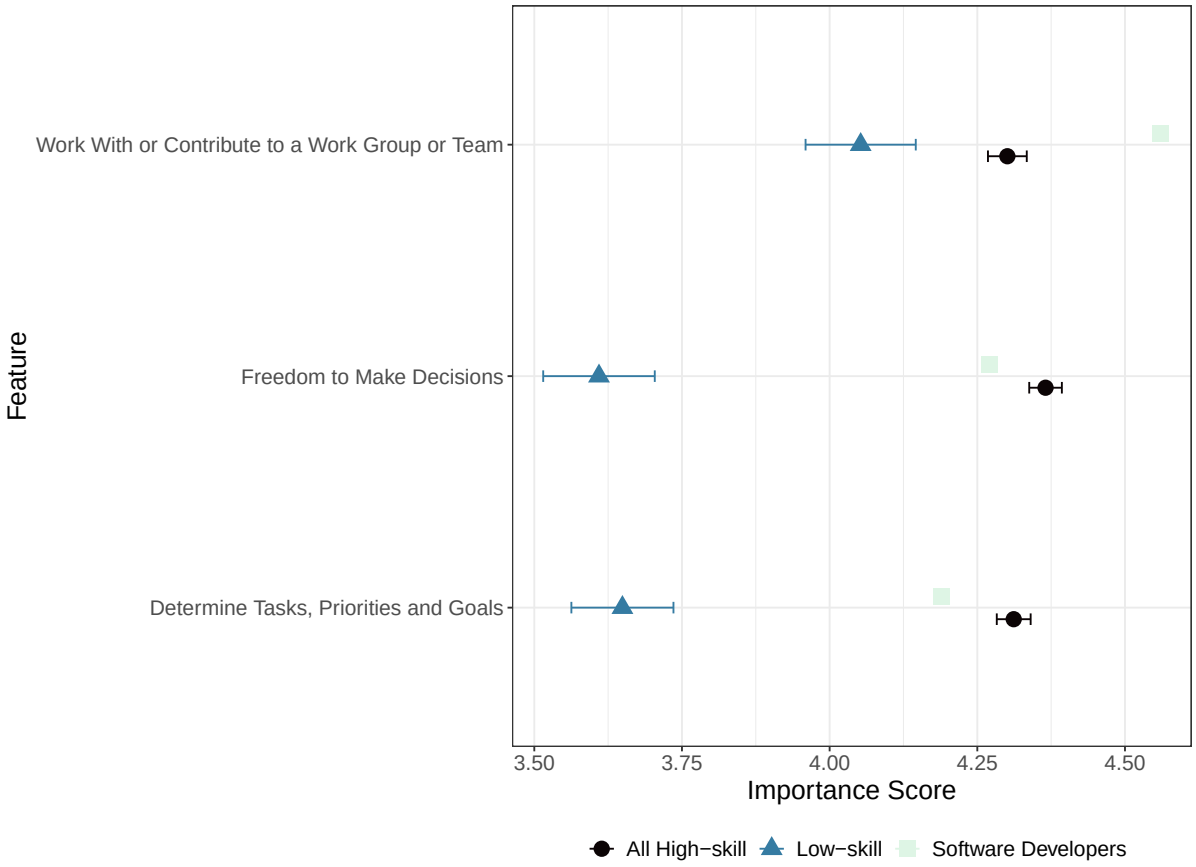
B Additional Figures

Figure B.1: Skill Requirements in High-Skill Occupations and Software Development



Notes: Based on data from O*NET Database Version 30.3. Transferable Skills Table. Light blue diamonds reflect average importance of the respective skill across all high-skill occupations, i.e. occupations that require at least a BA degree. Dark blue points reflect the importance of the respective skill among software developers, i.e. occupation 15-1252.00.

Figure B.2: Importance of Job characteristics among High-Skill Occupations, Software Development, and Low-Skill Occupations



Notes: Based on data from O*NET Database Version 30.3. Work Contexts Table. *All high-skilled* refers to occupations that require at least a BA degree. *Low-skilled* refers to occupations that do not require a high school degree. *Software developers* refers to occupation 15-1252.00. In the data, importance of a feature is measured on a scale from 1 (= not important at all) to 5 (= extremely important). In the plots, *high* importance aggregates across ratings 4 and 5, while *low* importance aggregates over values 3 and below. Figure shows means with 95% confidence intervals.

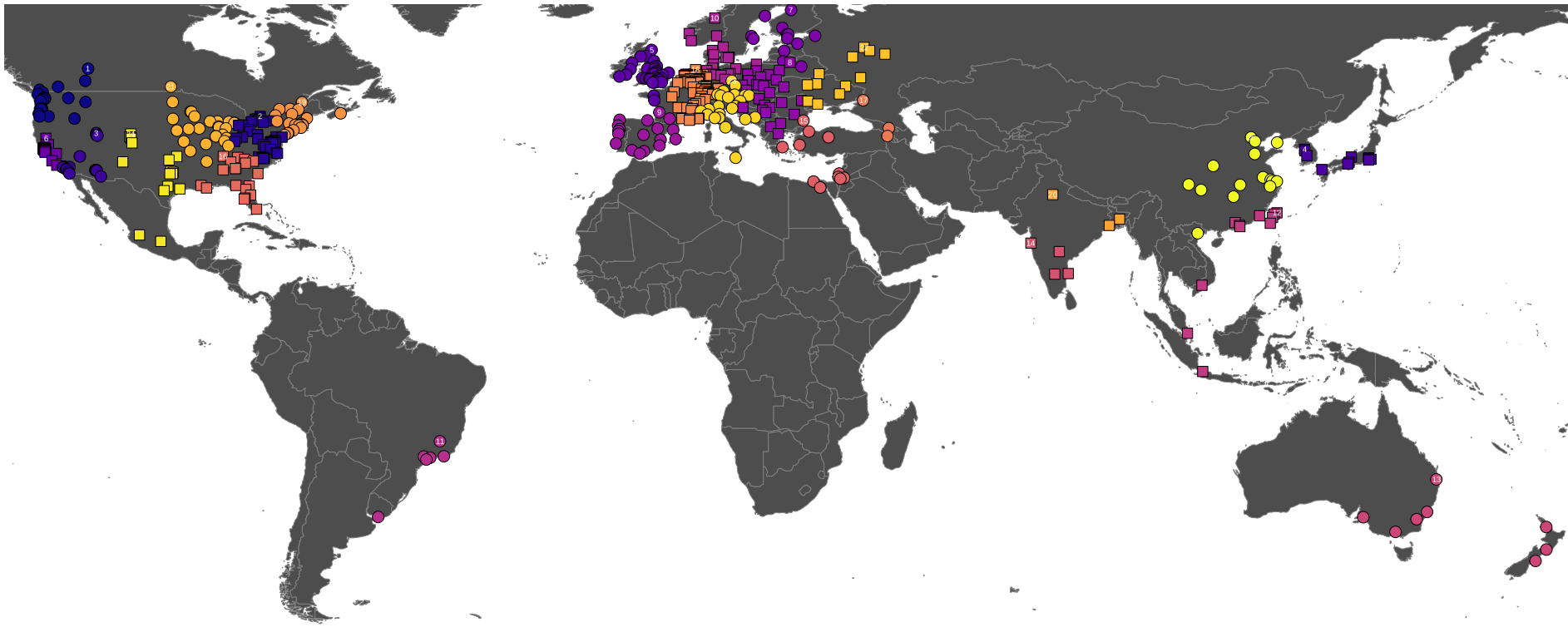
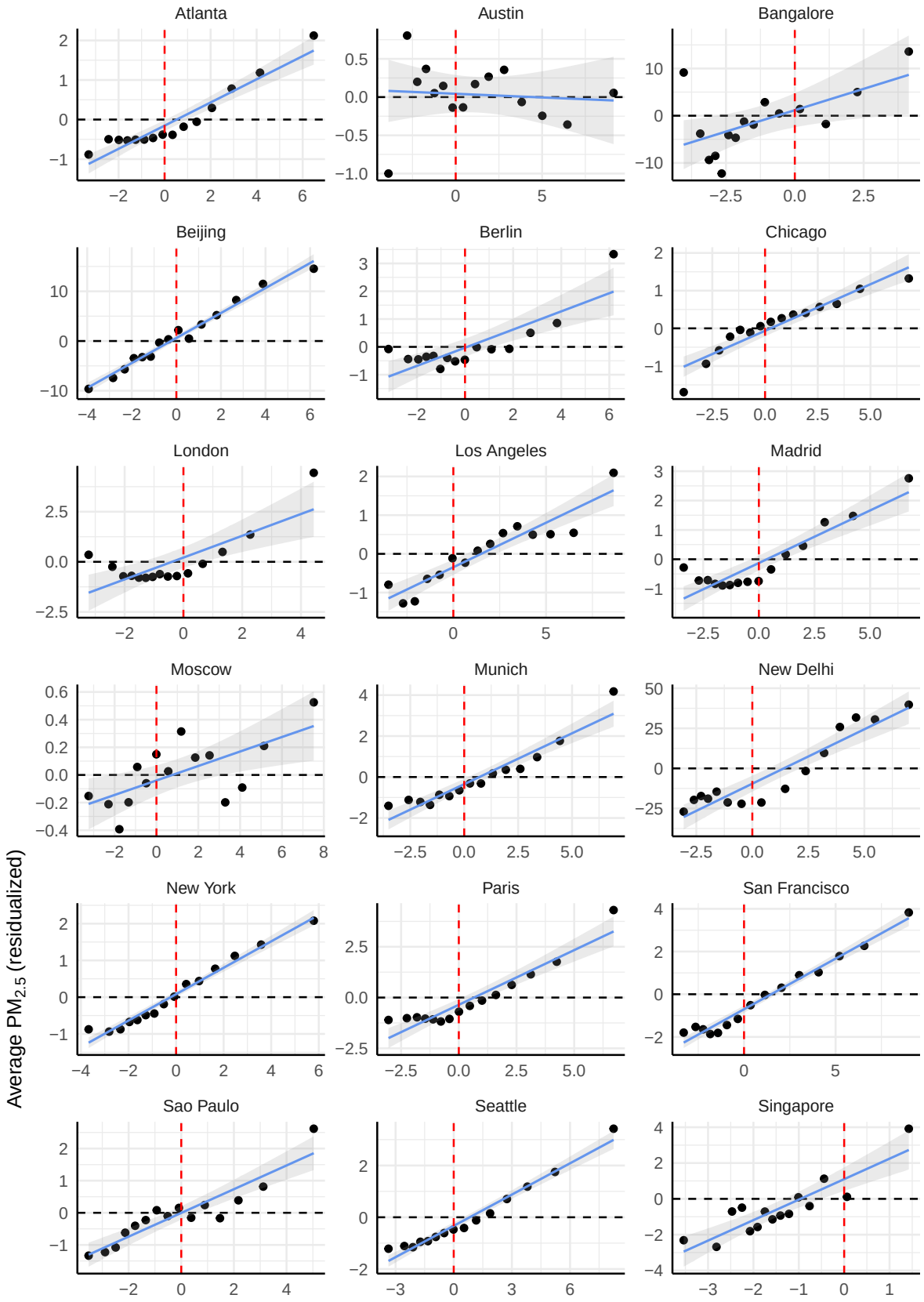
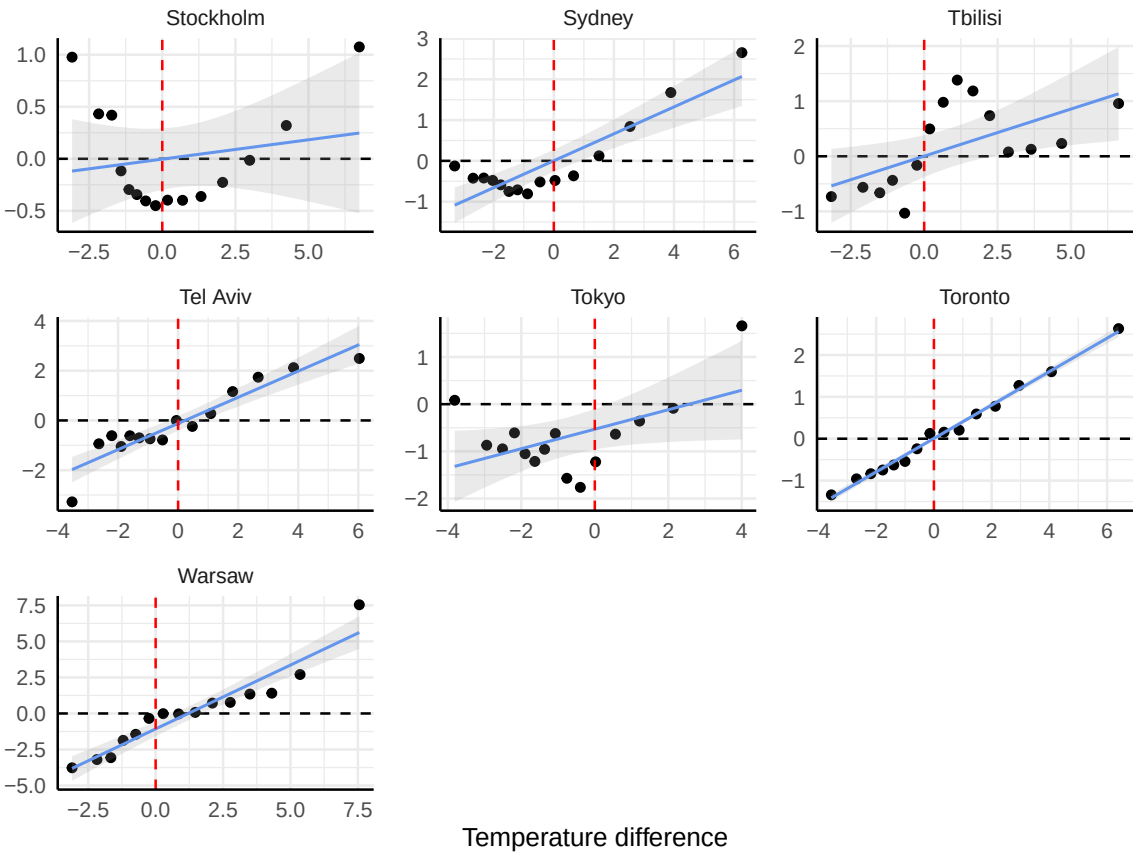


Figure B.3: Illustration of first stage city groups g

Notes: Cities with same color and marker are assigned to the same group g for the first stage regression (see Section 4 and Equation (2)).

Figure B.4: First stage for all 25 city groups





Notes: Binned scatter plots of residualized $PM_{2.5}$ concentration against inversion strength for each first stage city group. Residuals are generated by taking out weather controls and fixed effects as described in Equation 2. Blue lines show a linear fit with 95% confidence intervals. Plot titles denote one city from the respective group.

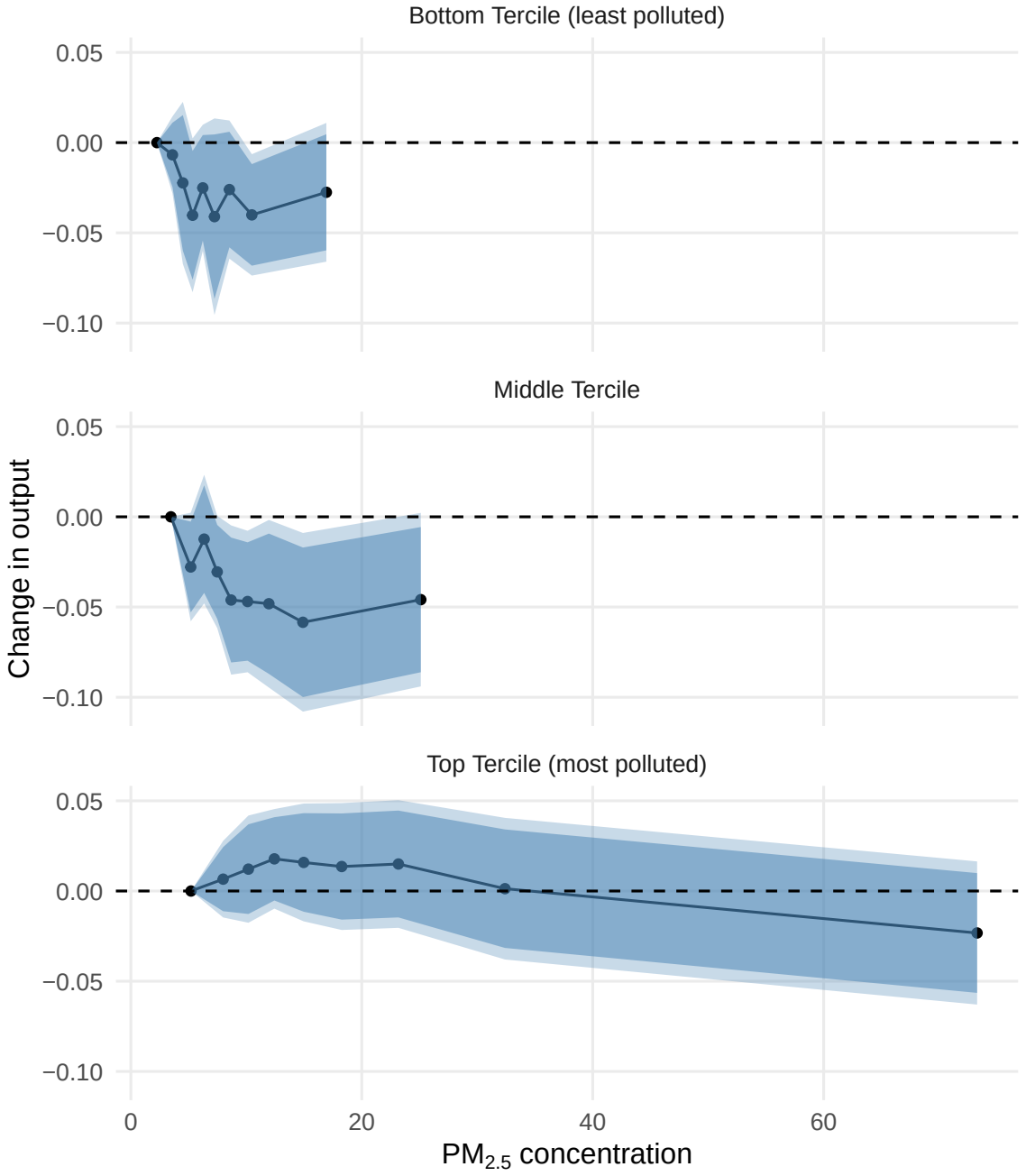


Figure B.5: Non-linear effects of $PM_{2.5}$ on Work Quantity across subsamples based on average $PM_{2.5}$

Notes: The figure depicts point estimates on different bins of $PM_{2.5}$ concentrations from OLS regressions of total actions on indicators for each bin for three distinct samples. Cities are assigned to subsamples based on average $PM_{2.5}$ concentration. Covariates: Weather and holiday controls as in Equation 1, region × date and city × month fixed effects. X-axis: Average $PM_{2.5}$ concentration in each bin in $\mu g/m^3$. Error bars indicate 95%- and 90%-confidence intervals.

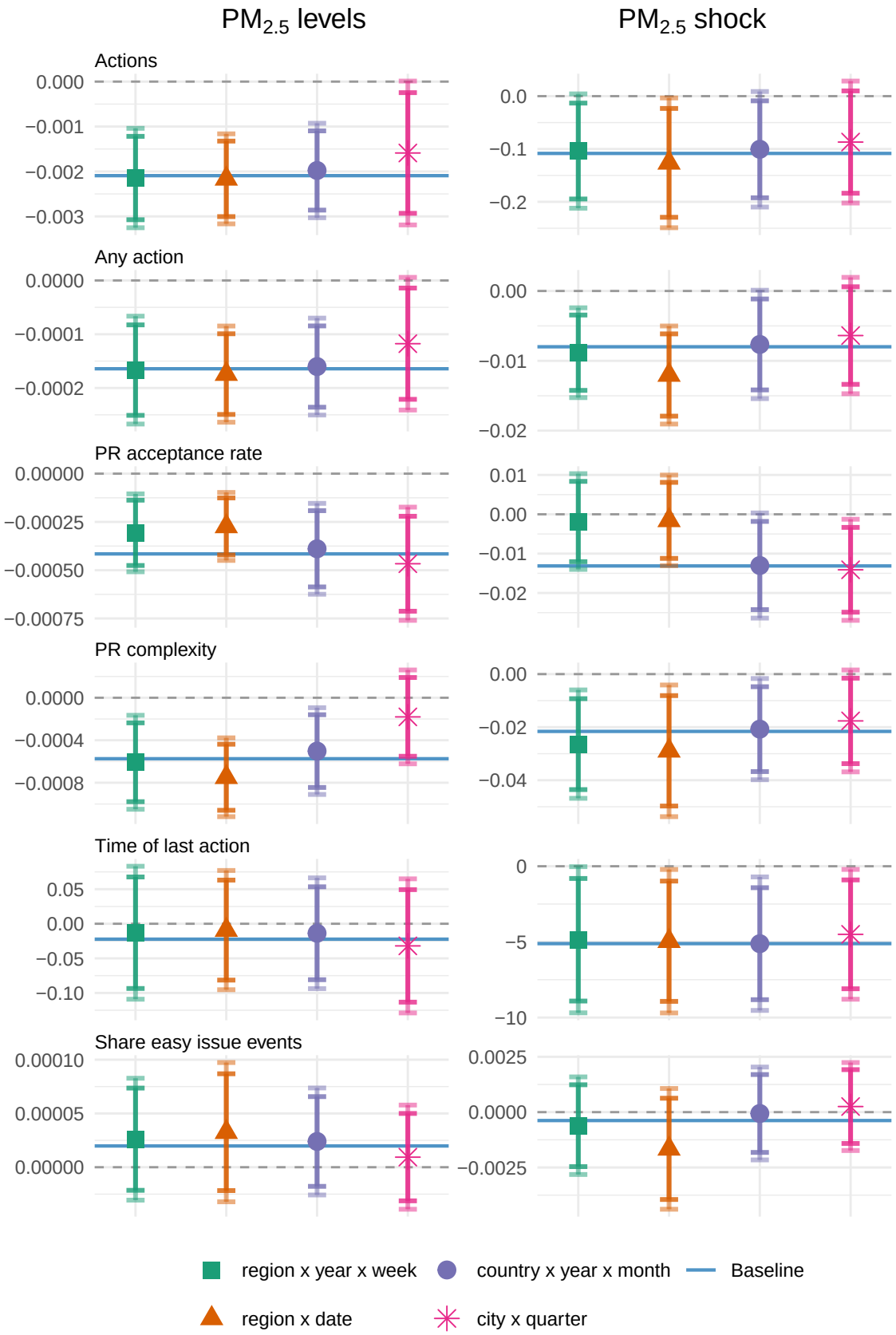


Figure B.6: Robustness to Changes in Fixed Effects

Notes: IV point estimates of the parameter β in Equation (1), along with 95% confidence intervals. Dependent variables are denoted at the top of each plot. Left side: regressor of interest is $PM_{2.5}$ concentration in $\mu g/m^3$. Right side: regressor of interest is an indicator for a $PM_{2.5}$ shock. Relative to specifications underlying results in Table 2, we change the included fixed effects, as stated in the legend. All regressions include control variables as described in Table 2 and are weighted by the number of active workers in a city during the current month. First Stage F-Statistics range from 71 to 512.

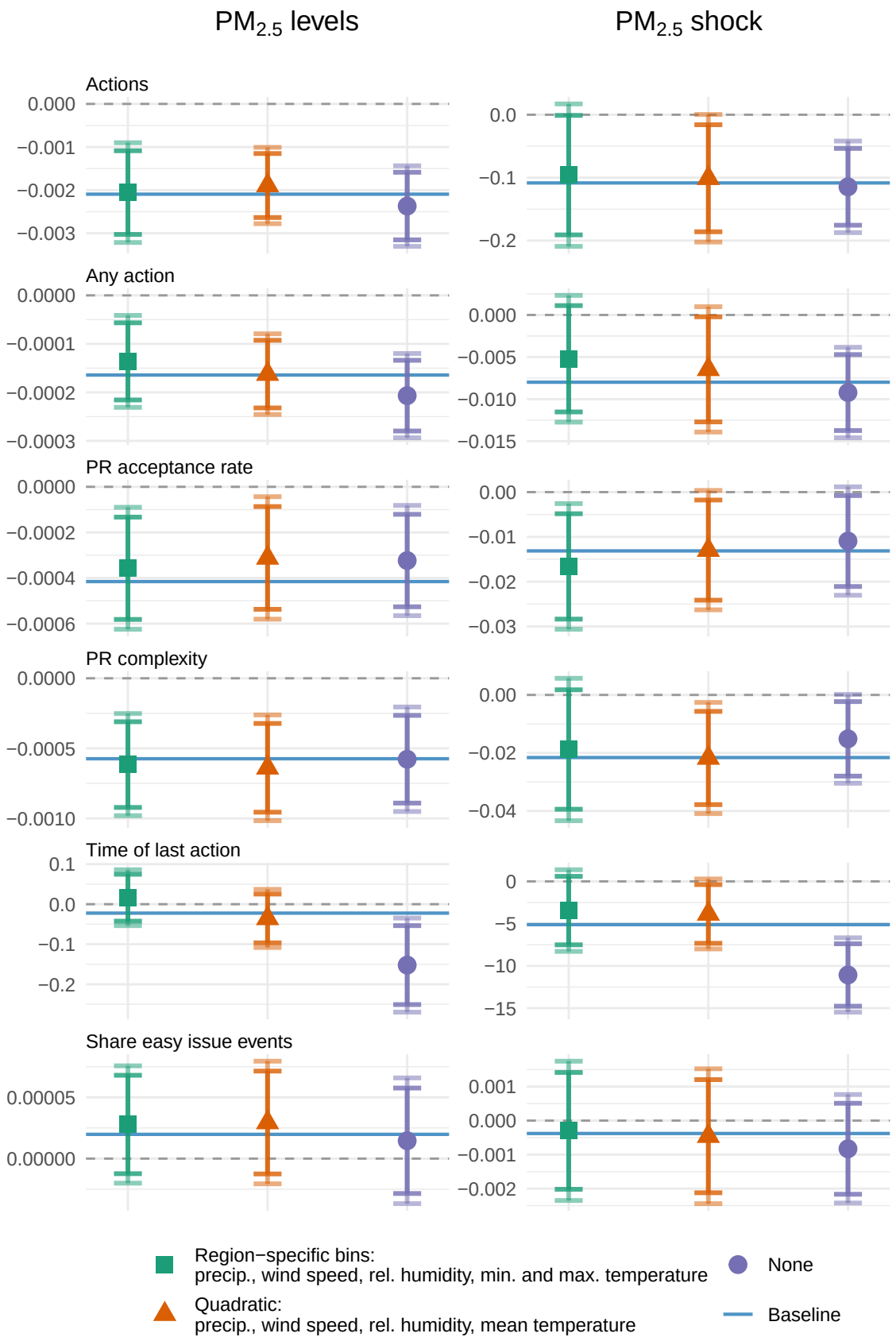


Figure B.7: Robustness to Changes in Weather Controls

Notes: IV point estimates of the parameter β in Equation (1), along with 95% confidence intervals. Dependent variables are denoted at the top of each plot. Left side: regressor of interest is $PM_{2.5}$ concentration in $\mu g/m^3$. Right side: regressor of interest is an indicator for a $PM_{2.5}$ shock. Relative to specifications underlying results in Table 2, we change the included covariates to control for weather conditions, as stated in the legend. All regressions include fixed effects as described in Table 2 and are weighted by the number of active workers in a city during the current month. First Stage F-Statistics range from 53 to 257.

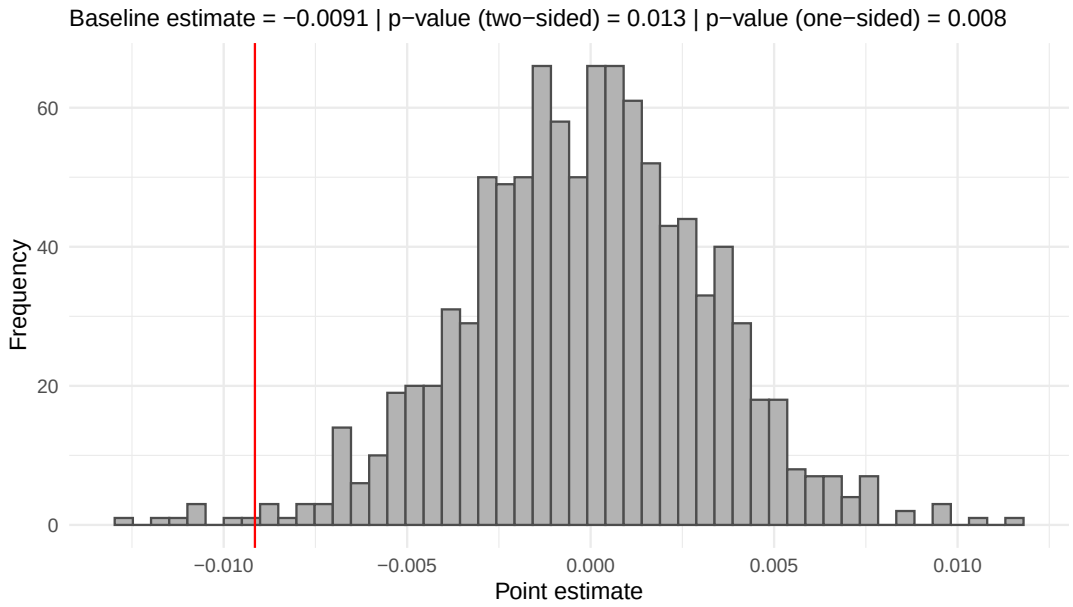


Figure B.8: Placebo Distribution of Long-Term Effects

Notes: The histogram shows the distribution of placebo coefficients from 1000 permutations that randomly reassign origin-city $PM_{2.5}$ across cities within move-year × continent groups. The red vertical line marks the actual baseline estimate using true $PM_{2.5}$.

C Gitcoin

This section provides additional details regarding the data collected from Gitcoin to assess the monetary value of output produced on GitHub and to validate some of our productivity and task complexity outcomes.

We collect data on 292 Gitcoin transactions via the Gitcoin API, including the type of the posted issue (bug, documentation, improvement, feature, or other), the expected issue difficulty as assessed by the issue funders (beginner, intermediate, or advanced), the URL to the PR solving the issue and awarded the payment, the value of the payment in USD, and the number of hours worked on the PR as stated by the PR author. The number of issues is relatively low compared to the volume of our GitHub data because Gitcoin is much younger than GitHub and only used by a small share of GitHub users. Using the URL of the PR, we combine this with information on pull request size obtained via the GitHub API, i.e. the number of commits it comprises, the number of lines of code added and deleted, and the number of files changed. This is possible because all Gitcoin issues and PRs are created in public GitHub repos and thus visible to us. In this context, a pull request reflects the complete work on a certain issue. Commits can be interpreted as single work steps in completing this task.

Combining the data on the amount of coding work done and on the payment made we can estimate the monetary value of output produced in public GitHub repos. The average monetary value per commit ranges from $32 in the subsample of issues of difficulty level *beginner* to $679 among issues marked as *advanced*. In the full sample, it amounts to $112. The mean time input per commit also exhibits a steep gradient with respect to difficulty: It is 1 hour at the beginner level, but 5.3 hours at the advanced level.

To validate the use of the number of commits per day as one of our core measures of developer productivity, we analyze how the number of commits in a PR correlates with the payment awarded and the time spent on it in the Gitcoin sample.

Table C.1 depicts results from regressions of the payment awarded for a PR, $log(payment_i)$, on the number of commits it comprises, $commits_i$ (columns 1-3), or the logarithm thereof (columns 4-6). We run specifications without any controls (columns 1 and 4), with controls for issue difficulty, issue type and the year of PR creation (columns 2 and 5), and alternatively with repository fixed effects (columns 3 and 6). The omitted difficulty category is *advanced*. Across specifications we find statistically significant positive effects, indicating that a higher number of commits is associated with higher payments. In terms of magnitude, the results from the regressions without any controls imply that one additional commit is associated with a 5.4% increase in payment (column 1), or that a 10% increase in the number of commits is correlated with a 3.5% rise in payment (column 4). When adding controls for issue difficulty and type, the magnitude of the effect is reduced. This reduction implies that part of the increase in payments in commits is driven by higher issue complexity. Even when using only variation across PRs submitted to the same repo, i.e., work on the same project, the positive relationship persists.

Table C.1: Validity Check: Number of Commits and Gitcoin Payments

	$\log(payment_i)$					
	(1)	(2)	(3)	(4)	(5)	(6)
$commits_i$	0.054	0.039	0.034			
	(.010)	(.009)	(.010)			
$\log(commits_i)$				0.348	0.264	0.192
				(.071)	(.068)	(.059)
$\mathbf{1}\{Difficulty_i = Beginner\}$		−2.399			−2.412	
		(.439)			(.419)	
$\mathbf{1}\{Difficulty_i = Intermediate\}$		−1.878			−1.851	
		(.415)			(.405)	
Year dummies		✓	✓		✓	✓
Issue difficulty		✓			✓	
Issue type		✓			✓	
Repository FE			✓			✓
Observations	292	274	292	292	274	292

Notes: The table presents results from OLS regressions using data on the sample of Gitcoin pull requests. Observations are at the pull request level. Dependent variable is the logarithm of the payment awarded to the PR author. Explanatory variables are the number of commits (columns 1 to 3) or the logarithm thereof (columns 4 to 6). Columns 2 and 5 add dummies for the year the pull request was created, dummies for issue difficulty, and dummies for issue type. Column 3 and 6 instead add dummies for the year the pull request was created and fixed effects for the repository. Robust standard errors are reported in parentheses.

In Table C.2 we present results from models where the dependent variable is $hoursworked_i$, the time input as reported by the PR author. We find that the time required to complete a task increases in the number of commits, and more so for issues of higher difficulty.

To validate our proxies for PR complexity, we run the specifications from columns 4 to 6 of Table C.1 again, but add the number of files changed in the PR and the logarithm of lines of code added as additional regressors. Results are presented in Table C.3. Holding the number of commits constant, adding more lines of code and changing more files is associated with a higher payment, suggesting that these variables indeed reflect task complexity.

Table C.2: Validity Check: Number of Commits and Hours Worked on a PR

	$hoursworked_i$			
	(1)	(2)	(3)	(4)
$commits_i$	0.375	0.939		
	(.132)	(.341)		
$\log(commits_i)$			2.375	10.346
			(.648)	(3.535)
$\times$ $\mathbf{1}\{Difficulty_i = Beginner\}$		−0.882		−9.748
		(.354)		(3.574)
$\times$ $\mathbf{1}\{Difficulty_i = Intermediate\}$		−0.667		−8.471
		(.349)		(3.560)
Observations	271	267	271	267

Notes: The table presents results from OLS regressions using data on the sample of Gitcoin pull requests. Observations are at the pull request level. Dependent variable is the number of hours worked reported by the PR author. In colunm 1 the only explanatory variable is the number of commits in the PR. Column 2 adds dummies for issue difficulty and interactions between the number of commits and the difficulty dummies. The ommited difficulty category is *advanced*. In columns 3 and 4 report results from the same models except that the number of commits is replaced by the logarithm thereof. Robust standard errors are reported in parentheses.

Table C.3: Validity check: PR complexity and Gitcoin payments

	$\log(payment_i)$			
	(1)	(2)	(3)	(4)
$\log(commits_i)$	0.143	0.136	0.070	0.145
	(0.067)	(0.068)	(0.058)	(0.056)
$fileschanged_i$	0.005	0.007	0.011	0.004
	(0.005)	(0.004)	(0.004)	(0.004)
$\log(linesadded_i)$	0.152	0.112	0.091	0.150
	(0.036)	(0.035)	(0.028)	(0.033)
$easylabel_i$				−0.348
				(0.173)
Year dummies	✓	✓	✓	✓
Issue difficulty dummies		✓		
Issue type dummies		✓		
Repository FE			✓	
Observations	292	274	292	270

Notes: The table presents results from OLS regressions using data on the sample of Gitcoin pull requests. Observations are at the pull request level. Dependent variable is the logarithm of the payment awarded to the PR author. Explanatory variables are the number of commits and the number of lines of code added in the PR (both in logs), the number of code files changed and dummies for the year the pull request was created. Column 2 adds dummies for issue difficulty and issue type. Column 3 instead adds fixed effects for the repository. Column 4 instead adds a dummy variable taking a value of one if the issue addressed by the PR carries a label that we classify as indicating an easy issue. The number of lines of code added and of files changed in the PR are winsorized at the 1st and the 99th percentile. Robust standard errors are reported in parentheses.

D Auxiliary Regressions

For estimating equation (1), measures of the output of each individual developer i are aggregated to the city-day level. Instead of forming simple averages, we take into account additional information at the developer level. This is done by estimating auxiliary regression, a common approach in this literature (e.g. Currie et al., 2015). In a first step, we estimate regressions for outcome y of developer i living in city c on day d of the following kind.

$$y_{i,c,d} = \mu_i + g(tenure_{i,c,d}) + \psi_{c,d} + \varepsilon_{i,c,d} \tag{1}$$

Here, $y_{i,c,d}$ denotes one of the measures of developer output, task choice, or working hours. The fixed effect μ_i captures time-invariant unobserved factors at the developer level. Including these is important as the composition of developers changes over time. A developer's experience is controlled for by $g(tenure_{i,c,d})$, a a cubic polynomial of time in days since registration on GitHub. Additionally, equation (1) includes city-day fixed effects. Their estimates $\widehat{\psi_{c,d}}$ give the average outcome for a city-day after controlling for experience and composition effects. These estimates replace the dependent variable in equation (1).

This approach is computationally less costly and asymptotically equivalent to directly estimating the regressions at the individual developer level (Donald and Lang, 2007). We take into account the sample variance by weighting all regressions by the number of underlying developer observations in each city-day cell (cf. Currie and Neidell, 2005; Isen et al., 2017).

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